

Contents

Volume 02 — Routing, Recon & Nessus (L11-L20)	2
L11: Routing Protocols — Part I	2
Learning objectives	2
What this lecture teaches	3
Key terms	8
Exam-oriented recap	9
L12: Routing Protocols — Part II	10
Learning objectives	10
What this lecture teaches	10
Key terms	17
Exam-oriented recap	17
L13: Routing Protocols — Part III	18
Learning objectives	18
What this lecture teaches	18
Key terms	23
Exam-oriented recap	25
L14: IP Version 6	25
Learning objectives	26
What this lecture teaches	26
Key terms	32
Exam-oriented recap	33
L15: Routing Examples	33
Learning objectives	34
What this lecture teaches	34
Key terms	39
Exam-oriented recap	39
L16: Demonstration — Part I	41
Learning objectives	41
What this lecture teaches	41
Tools mentioned	44
Exam-oriented recap	44
L17: Demonstration — Part II	45
Learning objectives	45
What this lecture teaches	45
Tools mentioned	50
Commands / syntax from the lecture	50
Key terms	50
Exam-oriented recap	50
L18: Demonstration — Part III	50

Learning objectives	50
What this lecture teaches	51
Tools mentioned	56
Commands / syntax from the lecture	56
Key terms	58
Exam-oriented recap	58
L19: Nessus Installation	58
Learning objectives	58
What this lecture teaches	59
Tools mentioned	60
Exam-oriented recap	60
L20: How to Use Nessus	60
Learning objectives	60
What this lecture teaches	60
Tools mentioned	63
Exam-oriented recap	63

Volume 02 — Routing, Recon & Nessus (L11-L20)

Course: NPTEL Ethical Hacking (106105217) · **Instructor:** Prof. Indranil Sengupta, IIT Kharagpur

Series: notes-2 (transcript-grounded) — Routing protocols, IPv6, live OSINT/Nmap demos, Nessus vulnerability scanning.

Study notes derived from YouTube/Digimat lecture captions. For authorized lab use and exam preparation only.

L11: Routing Protocols — Part I

Transcript: [11-routing-protocols-part-i.md](#)

Video: <https://www.youtube.com/watch?v=zxToIFJFkMw>

Week / theme: Week 3 — Routing protocols, IPv6

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- So, this is what we shall be starting our discussion with in this lecture.
- So, let us look at broadly the Internet routing protocols, how they work?.
- Now, exactly how this routing or forwarding of the packet happens, these are dependent on something called routing protocols or routing algorithms.
- There are networking devices as we had mentioned at the beginning called a routers which are primarily responsible for forwarding or routing of these IP packets ok.

- So, this is what we shall be starting our discussion with in this lecture.

What this lecture teaches

Core concepts from the lecture

If you recall our discussion so far we have talked about the TCP/IP protocol suit, where if you recall, we have seen how packets flow through the Internet, through the network via the IP layer in TCP/IP, so that the packet can be forwarded or routed correctly from any source to any particular destination.

Now, exactly how this routing or forwarding of the packet happens, these are dependent on something called routing protocols or routing algorithms.

There are networking devices as we had mentioned at the beginning called a routers which are primarily responsible for forwarding or routing of these IP packets ok.

So, this is what we shall be starting our discussion with in this lecture.

The lecture is titled Routing Protocols, the first part.

Now, in this lecture we shall be broadly looking at two things; first we shall be talking about the different packet delivery options which are there, when a packet flows through the Internet and some of the alternate routing techniques or routing methods that are possible in this context ok.

So, let us look at broadly the Internet routing protocols, how they work?

Before going into the routing protocols, let us talk about something called connection options.

Continued

Now here what we are talking about is, suppose I have a source node which is trying to send some packet to a destination node.

These may be computers connected to the Internet and there can be various intermediate nodes which in this context a routers through which the packets may flow ok.

The source may be connected to one or more such routers and so, here I have shown two connections.

For example, there can be a multiple such connections right.

Now, you recall when we talked about TCP/IP protocol suit, we talked about the two different protocols that are mostly used at the transport layer level namely TCP and the UDP.

TCP is connection oriented UDP is connectionless.

So, what it really means, let us again recapitulate connection oriented means, the network protocol whatever protocol you are using at the transport layer and the network layer level, must first be establishing so called connection.

In TCP there is a connection establishment phase, we talked about using three way handshake ok.

Continued

Now, once connection is made, a pure connection oriented protocol says all packets are delivered as per the connection ok.

You see here there is a catch, this statement does not mean that all packets will be following the same path.

Connection is a logical concept.

When I say that I am connected with a destination, this means when I will be sending messages, receiving messages, I will be keeping track of my connection; that means, how many bytes have been sent; how many bytes are remaining to be sent?

Similarly, for the other side also.

So, it does not mean that when I am sending the bytes, they all will follow the same path.

This would happen of course, when you are having something like circuit switching.

In circuit switching you recall there all data will be following the same path.

Continued

But when you are talking about a packet switch network like TCP/IP, there it does not happen like that.

A logical connection is maintained at the TCP level, but at the IP, the packets can follow any available route.

So, this is what is meant by connection oriented and connectionless just like UDP or IP, connectionless means there is no need for a prior connection establishment.

The data packets that you are sending, they are sent as independent entities.

They will be routed through the network without any context in an independent way and they will finally, reach the destination.

Now, we mentioned that this UDP protocol which is a classical connectionless protocol, which is used at the TCP level, does not guarantee reliability in packet transfer.

So, if you need reliability and if you are using UDP, you have to take care of error correction and this checking explicitly at your application layer level ok, fine.

So, this is what I mentioned TCP and UDP are protocols at the transport layer level, but below it you have the IP protocol which runs at the network layer level which is essentially a connectionless packet delivery system, IP.

Continued

So, even if at the higher layer TCP talks about connection oriented, but when the packets are given to IP for actual delivery, IP can follow any path, arbitrary paths.

So, this actually means that in the TCP protocol when the packets are sent between a source and a destination, packets may follow different paths.

But the TCP protocol maintain some information about the connection such that the application has a feeling that well there is a connection, there is a reliable mode of connection, no errors are there, no bits are getting lost and data are being received exactly in order ok.

This is what TCP provides over IP.

Now talking about the packet delivery options, I am particularly talking about the IP layer here, because as I said, TCP/IP is the most dominant protocol that drives the Internet and IP is the most widely used protocol at the network layer level which is responsible for packet routing.

So, when you talk about packet delivery options, we are talking about a scenario like this as I have shown in this diagram.

There is a network as you can see in the middle, there is a network, there can be some computers which are directly connected to the network.

You already seen, you have seen networks in your organization, in your office wherever you are.

Continued

There are computer networks, there are networking switches or hubs and you have connection to your computer through these networking devices right.

So, this hosts can have direct connection to the network and also there can be some routers; routers can also have connections to this network.

Now, the question is, in this network there can be multiple hosts H1, H2, two I have shown.

There can be several others.

So, all these hosts can communicate among themselves through this network.

So, the question arises why do we need this router?

This router is required when there is a communication need with other networks, whenever you want to communicate with the outside world.

Well, either one of this hosts is trying to send a packet to some host which is outside this network or some incoming packet is coming from somewhere which is destined to one of these hosts.

Continued

Now the first mode is called a direct delivery, where we are not talking about the outside world.

We are talking about a scenario like this where this H1 can directly send a packet to host H2 via this network, because both of these are connected to the network, they have a connection.

So, they can directly send a packet or receive a packet.

This is called host-to-host.

And the other can be host-to-router or router-to-host, there can be one or more routers.

One I have shown, there can be more routers connected, there can be other routers also connected.

For example, there is another router here.

So, router-to-host means suppose a packet has come to the router, the router can send this packet to a particular designated host or the other way around, host wants to send a packet to the router.

Continued

Now, this direct delivery option is possible when the entities which are communicating, they are all connected to the same network.

This is the constraint ok, this you should remember.

Then comes indirect delivery.

Indirect delivery means the source and destination are not part of the same network.

They belong to two different networks.

So, as this diagram shows, let us suppose this H1 wants to send a packet to host H2.

So, as you can see, there are three networks.

Here, this N means networks.

Continued

There are three networks, but I have shown only some, there can be other connections also to the network, other hosts and routers.

There can be connections ok.

So, I have shown only few.

Now, when this host wants to send data to H2, as you can clearly see, they do not belong to the same network.

So, data cannot be sent directly.

So, what will happen?

Somehow H1 will decide that the packet that I am sending has to be sent to R1.

So, it will first send the packet to router R1.

Continued

The R1 can again have other connections, but R1 will decide depending on the destination address that this packet has to be forwarded via this network, middle network to another router R2 right and R2 will finally, come to know that well destination is in the same network where I am.

So, I can directly send the packet to H2.

So, the last phase, it is direct delivery; in the first two steps, it is indirect delivery right.

This is how packet forwarding and delivery can happen.

Now, talking about routing method, you see each entity in the network, routers of course, but even the computers, they have some kind of routing information maintained, that is called a routing table.

Routing table will tell you that well if I want to send a packet to some other computer whose IP address I know, where to send that packet.

Well, a router obviously, will be having multiple connection.

So, there is a decision, but even for a computer, it is possible for you to have multiple network cards and multiple network connections with your computer.

Continued

So, your computer can also decide where to forward the packet.

So, in a sense you can even make a normal computer act like a simple router.

It can also take some routing decision, the IP layer of it.

Now, talking about the routing methods broadly speaking, the routing table contains various different kinds of entries, as you shall see through examples.

They can be something called next-hop routing, network-specific routing, host-specific routing and default routing.

Next-hop routing means I tell that to reach a particular host which is the next network or next router to follow that is called hop.

One router to the next router, this is defined as a hop.

So, when a packet moves there will be multiple such hops, till it reaches the destination network where the packet can be delivered directly ok.

Continued

Network-specific routing means there are some situations where you may want to transmit a packet to all hosts of a particular network.

So, there you are not sending the packet to a particular IP address, but to a network as a whole.

This is network-specific routing, you are specifying a network address as the destination.

Similarly, you can specify a particular host that is called host-specific routing where to follow and if nothing matches in the routing table, there will be a default entry.

If nothing matches, you take the default route.

This is how typically routing tables are organized and these are the kind of entries which can be there fine.

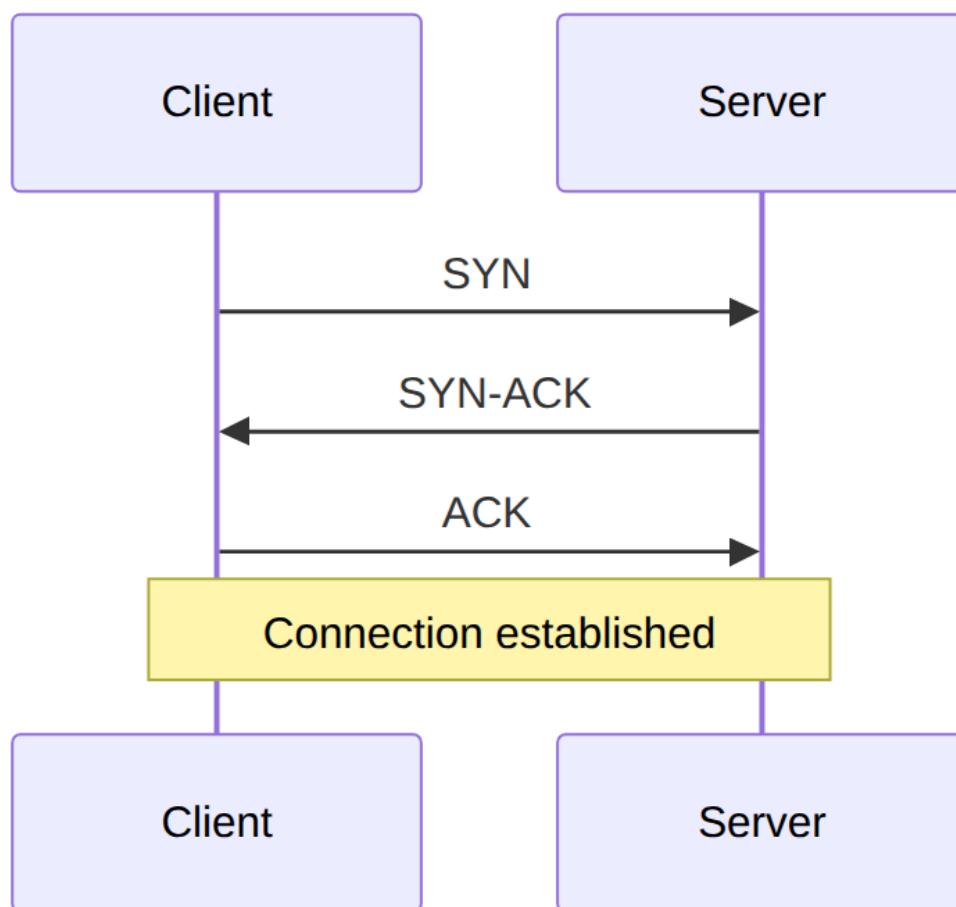


Figure 1: Architecture diagram

Key terms

Term	Meaning in this lecture
When I say that I am connected with a destination, this	when I will be sending messages, receiving messages, I will be keeping track of my connection; that means, how many bytes have been sent; how many bytes are remaining to be sent?
So, this actually	that in the TCP protocol when the packets are sent between a source and a destination, packets may follow different paths.
Now the first mode	a direct delivery, where we are not talking about the outside world.
This	host-to-host.
So, router-to-host	suppose a packet has come to the router, the router can send this packet to a particular designated host or the other way around, host wants to send a packet to the router.
Indirect delivery	the source and destination are not part of the same network.
Here, this N	networks.
Next-hop routing	I tell that to reach a particular host which is the next network or next router to follow that is called hop.
One router to the next router, this	a hop.
Network-specific routing	there are some situations where you may want to transmit a packet to all hosts of a particular network.
Similarly, you can specify a particular host that	host-specific routing where to follow and if nothing matches in the routing table, there will be a default entry.
And lastly, host-specific routing	in the destination I can directly specify a host and this host can be connected to the same network also.

Exam-oriented recap

- So, what it really means, let us again recapitulate connection oriented means, the network protocol whatever protocol you are using at the transport layer and the network layer level, must first be establishing so called connection.
- This router is required when there is a communication need with other networks, whenever you want to communicate with the outside world.
- This is the constraint ok, this you should remember.
- Now, lastly I said whenever nothing matches, you have to follow a default route like you take an example like this, where I have a host H1 here.
- So, you can make some entries in the routing table which will never change that is fixed.
- So, when with time whenever packets are flowing, there can be some dynamic behavior that may happened like some link might go down, some host might be come down.

- So, whenever an address comes, you do a bit by bit AND with this and then, you compare whether it is matching with this.
-

L12: Routing Protocols — Part II

Transcript: [12-routing-protocols-part-ii.md](#)

Video: <https://www.youtube.com/watch?v=hiEeGehQs3g>

Week / theme: Week 3 — Routing protocols, IPv6

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- Let us continue with our the discussion on Routing Protocols which we had initiated during the last lecture.
- So, we continue with the discussion.
- You see by the very name interior, exterior, you can understand, we are talking about some kind of a boundary that whether we are inside the boundary or outside the boundary ok.

What this lecture teaches

Core concepts from the lecture

Let us continue with our the discussion on Routing Protocols which we had initiated during the last lecture.

In the last lecture, we talked about routing tables, the way packets get delivered and some typical fields or entries that are present in the routing table.

So, we continue with the discussion.

The topic of this lecture is Routing Protocol Part II.

In this lecture, we shall be first distinguishing between two different kinds of routing protocols called interior and exterior and specifically, we shall be briefly looking at two different practical routing protocols; one is called Routing Information Protocol or RIP, other is Open Shortest Path First or OSPF.

First as I have said we would be distinguishing between two different classes of routing protocols; one is called interior, one is called exterior.

You see by the very name interior, exterior, you can understand, we are talking about some kind of a boundary that whether we are inside the boundary or outside the boundary ok.

Now, interior means see these protocols are used for routers to communicate among themselves.

Continued

Multiple routers they communicate using these protocols for updating their routing tables.

So, we saw in the previous lecture that there is a field in the routing table which indicates some flags that whether a route was added or was modified through redirection.

So, these protocols help in this redirection and can automatically make updations in the routing table entries right.

Now, interior protocols for which examples are RIP and OSPF, there the idea is that the routers that are talking among themselves for updating their routing tables, they are all inside some kind of a boundary ok.

They are all inside, that is why they are called interior and exterior means well let us say there is another boundary.

There is some other router there and one of these routers can talk to that other router in that other boundary and update routing information, share routing information accordingly.

These are called exterior routing protocols and border gateway protocol is one very important example of that.

Now, that boundary I was talking about, that boundary technically is known as an autonomous system ok.

Continued

Autonomous system is that imaginary boundary which I am, I have just mentioned, so, that imaginary boundary, an autonomous system or in short AS, this is an AS and how do we define an AS?

AS is loosely defined as inside this, there can be multiple routers, there can be many routers, there can be many hosts, many networks also, there can be multiple networks not a single network ok, multiple networks and obviously, there are large number of computers, hosts.

Now, if all of them are managed by a single organization, then we can define it as an AS.

Many organizations may have different sections or departments, there can be multiple networks connected together, but they are all managed by a single entity or organization.

So, if the organization wants they can define their organization as an autonomous system or AS.

There is a procedure you can get a unique autonomous system number for your network and all autonomous systems in the world, their number, numbering is unique; no two autonomous systems have the same number ok.

Now, inside this autonomous system, the routers within the autonomous system, they can exchange information using some common routing protocol.

Because there inside the organization, you are following the same rules, same protocols and AS graph means graph is what, some nodes connected by some edges.

Continued

So, in some means in a network graph, we have seen some examples.

The nodes are the hosts, routers and networks; edges are their interconnections.

So, if the graph is connected which of course it has to be, because if there is no path between 2 nodes, then you really cannot reach there.

The graph is disconnected, but if there is some failure, some link failure; then, your graph might get disconnected ok.

But normally, the AS graph is connected and as I had said every autonomous system, if you register with a central authority, you will be assigned a unique AS number and routing protocols between routers belonging to the same AS will be following interior routing protocols and across different AS's, they will be following exterior routing protocols.

This is the difference.

Now, here I am showing a picture, where there are three autonomous systems, I am showing those numbers are 1, 2 and 3.

Now, as you can see in each autonomous system, there are several routers, there are several networks.

Continued

Of course, there are computers.

I have not shown the computers here and there are links which connect routers to the networks.

So, the router which have there inside, you can see that there are multiple connection for AS-1 for example, this router is connected to this router via this network.

This router has a connection to this router via this network and this router has a connection here.

Now, these routers can exchange information and keep their routing table updated using interior protocols which I said, there are two main examples RIP and OSPF.

But when you talk about routers which belong to different autonomous systems which are shown by these blue lines, here we use some external or exterior routing protocol like BGP ok.

This is how these interior and the exterior protocols work and this is how their scopes are defined all right.

This already I have mentioned, I am just repeating once more that which class of protocol you need to use for a particular scenario.

Continued

Well, for routers within an autonomous system, inside the autonomous system, you use interior routing protocols RIP or OSPF and to exchange routing information between routers in different autonomous systems, you use exterior routing protocol like BGP.

Now, let us briefly look at these routing protocols one by one.

First, we talked about this RIP, Routing Information Protocol.

Now, I mentioned already, it is an interior routing protocol meant to be used inside an autonomous system.

If this is an autonomous system, routers that are present inside this AS, they will be using RIP to exchange information for updating the routing tables.

Now, you may ask why this exchange is required because they are anyway belonging to the same organization.

But you see, consider an organization where there can be two departments, two different networks.

There is one router sitting here; one router sitting there.

Continued

The other router will know much better about the status of the other network whether some link is down or everything is alright or not.

But when I am sitting in this network, I want to send a packet to one of the computers in the other network, I do not know what is the status in the other network; which path is this best; which path is required to be followed.

So, the other network can give me some information about the current state of the network.

Current state; what is the links that are currently down; what are the links that are currently congested; various information like that.

So, I can update my routing table so that I can choose the best route to follow to reach that host ok, all right.

Now, in RIP the way the routing tables are updated, there is a very standard method called distance vector routing.

Well, I am not going into the detail of these methods.

These methods are given in significant detail in any standard textbook on computer networks, you can see through them if you are interested.

Continued

Distance vector routing basically says, every router will have some information about how far the other routers are from myself.

Like for example, I have a router here; there can be other routers like this.

Let us say, this is router 1, router 2, router 3 and router 4.

Let us suppose the connections are like this, router 1 will have the information that router 4 is 3 hops away from me ok, but suppose a new link gets established between R1 and R4, so that 3 will get updated to 1.

Now, R1 and R4 are directly connected.

Similarly, if one of the links go down, these values will change ok.

Distance vector routing sends the distances to all routers whatever information I have to all the other routers and everyone collects all the information they receive and in a consolidated way, they update their routing tables.

This is what is done here and some timers are used to detect link failures.

Continued

If some packet you are sending, you are not getting acknowledgement within certain amount of time, then you may assume that the link has failed; link is not working.

This was used in older networks, but presently this RIP is very rarely used because RIP has some drawbacks.

Some of the drawback is that the way the routing tables are updated using distance vectors, see once distance vectors are shared, something will change.

That change again will be shared that will again trigger some more change.

So, this leads to something called convergence in the routing table which at times is very slow.

Suppose a link fails, again a link goes up.

So, how quickly can the routers respond to these changes and update their routing table.

This RIP is not very good at that.

Continued

It can take a lot of packet exchanges to update all the routing table to reflect the correct status.

So, broadly speaking I am just mentioned here if a network becomes inaccessible due to a link failure, some link has failed; it may take a long time for all other routing tables to know about this.

Because you see I am giving a very small example.

Let us say R1 is there, R2 is there, R3 is there and let us say R4 is there.

R1, R2 this is the connection.

Now, let us say I am thinking about the distance to R4.

Well, R1 knows that R4 is 3 hops away; R2 knows R4 is 3 hops away; R3 knows R4 is 1 hop away and R4 is R4, it is already R4, 0 hop.

Now, suppose this link has failed and let us say there is another router here, let us say R0 which initially was 4 hops away, this was 2 ok.

Continued

Now this link has failed.

So, you see now R1 has no path to R4, the network has become disconnected; but R0 will tell R1 that well R4 is 4 distance away from me and because this is 1 hop.

So, R1 will update this to 5; 4 plus 1.

But actually there is no path.

So, in this way this 5, then 6, then 7, it will go up slowly, this is sometimes called counting to infinity; counting to infinity problem and this becomes very slow and too much bandwidth gets consumed due to the routing updates.

This is some drawback here; some of the drawbacks.

Now, this OSPF, Open Shortest Path First, this is more widely used as the interior routing protocol.

Let us see some of the features of OSPF.

Continued

Now, in TCP/IP networks, this is most widely used as I said and it relies on something called link state advertisements.

Well, here distance vectors are not shared like how far R1 is from me, R2 is from me, R3 is from me, nothing like that, but status of the links, some link what is the current delay from myself.

It sends or shares the link delays, if a link is down, it tells the link is down.

But in the earlier case that information was never sent ok.

Basic concept is that the OSPF tries to compute a route again dynamically based on some least cost algorithm ok.

The notion of cost again can be configured how, means how do we define cost.

Is it minimum delay; is it minimum cost or is it with respect to the data rate of the links you are following ok; how are you defining?

So, each router will be maintaining the information about the current state of the network, of the autonomous system that we are calling as the database.

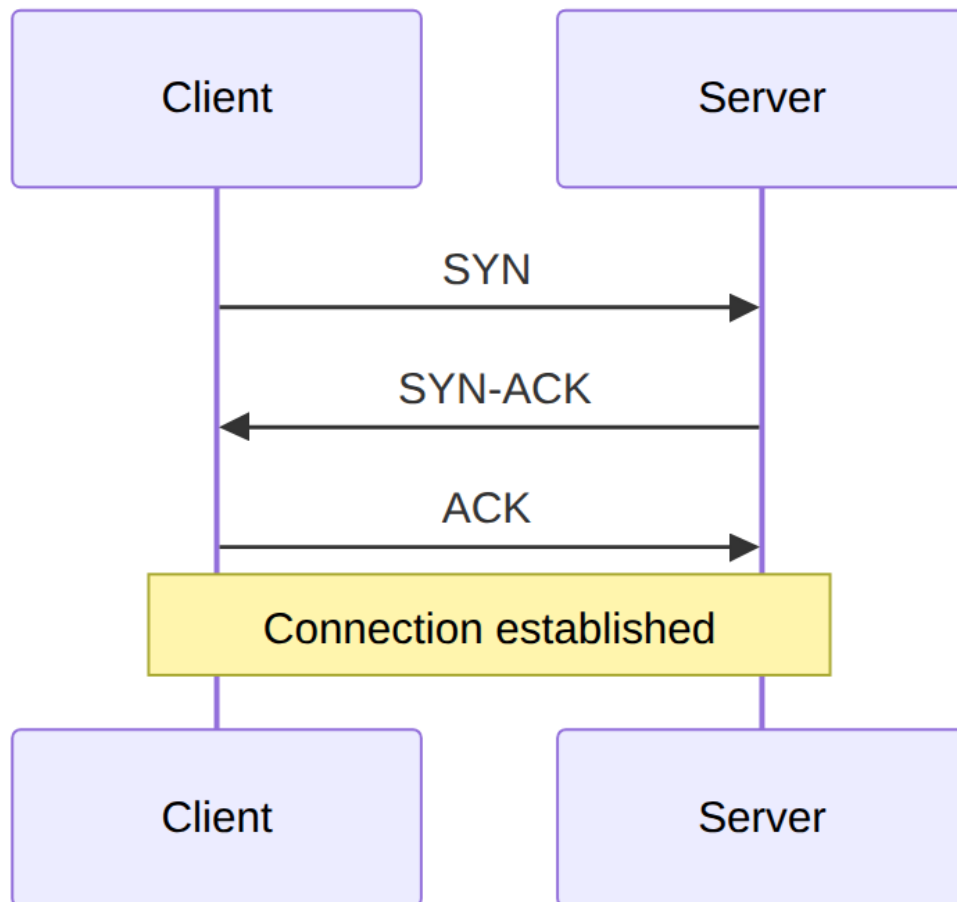


Figure 2: Architecture diagram

Key terms

Term	Meaning in this lecture
Now, interior	see these protocols are used for routers to communicate among themselves.
So, in some	in a network graph, we have seen some examples.
The notion of cost again can be configured how,	how do we define cost.
This	weight, now there can be two kinds of edges; one is an edge between two routers.
That is why some authentication, authentication	I must be sure that the link state advertisement is coming from the right person, right router and some authentication information and of course, the actual link state advertisement or whatever you are sending, they actual routing data, link states etc.
This	link state request and that link state advertisement is this link state update, that is actually the link state information which a router sends to the some other router and link state acknowledgement means once some router receives a link state advertisement, it will send back an acknowledgement that well I have received it correctly, so that other routers know that what I have sent was received correctly by everyone else right and regarding authentication type I said the simplest case may be clear text.

Exam-oriented recap

- These are called exterior routing protocols and border gateway protocol is one very important example of that.
- But in the earlier case that information was never sent ok.
- Well, if someone maliciously sends a wrong link state advertisement, then all the routing table might get updated in the wrong way and packet might follow some circular loops and never get forwarded to the right direction.
- It might specifically request link state information from some other router.
- Everyone can see what is going or you can encrypt it, so that if someone wants to hack my network and make changes in my routing table by altering these packets, it will be difficult to do so ok.
- These are the options which are available.
- So, with this I come to the end of this lecture where I have very briefly talked about the Interior Routing Protocols, RIP and OSPF.

L13: Routing Protocols — Part III

Transcript: [13-routing-protocols-part-iii.md](#)

Video: https://www.youtube.com/watch?v=_XWiJpLE4OM

Week / theme: Week 3 — Routing protocols, IPv6

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- In this lecture, we continue with our discussion on Routing Protocols.
- So, the topic of today's discussion is Border Gateway Protocol and some of the message types in BGP.
- So, talking about the Border Gateway Protocol, BGP, well as you can understand from the name we are talking about something called border.
- Now, exterior and interior, I explained the distinction earlier.
- So, once the other router sends some information to this particular router; so we will have some information about the other place, how the network situation is; what are the available routes and so on.

What this lecture teaches

Core concepts from the lecture

In this lecture, we continue with our discussion on Routing Protocols.

If you recall in the last lecture, we discussed the two interior routing protocols namely the RIP and OSPF and in this lecture, we shall be talking about the exterior routing protocol that is most widely used in the internet with the help of TCP/IP that is BGP.

So, the topic of today's discussion is Border Gateway Protocol and some of the message types in BGP.

I shall also show you some very overall high level example that how BGP works, just to give you a conceptual idea ok.

So, talking about the Border Gateway Protocol, BGP, well as you can understand from the name we are talking about something called border.

You see for every country, border is an interface with some other country right; similarly in a network when you say a border, border gateway it means; it is like a router which is often called a gateway which connects this network to the outside world to other networks, to other you can say router ok.

So, let us see what BGP exactly is.

Now, as I said BGP is the most widely used exterior router protocol.

Continued

Now, exterior and interior, I explained the distinction earlier.

Let us say when you have several autonomous system.

Suppose, these are all autonomous systems, which consists of networks and routers. So, you can have one router here, one router here and may be one router here, these are exterior routers or these are border routers.

These border routers can connect with each other depending on means how you want to connect, they are going to be multiple ways of connection and it is over this links that connect the border routers you have this BGP protocol running.

So, you will also have BGP here, but inside the autonomous systems will be having some of the interior router protocols running ok.

Now, what is the role of BGP?

BGP allows these routers to exchange routing information.

Continued

Like for example; the router sitting in this particular autonomous system does not know what is the network status of the other two networks, other two autonomous systems.

So, once the other router sends some information to this particular router; so we will have some information about the other place, how the network situation is; what are the available routes and so on.

So, that the local routing tables can be updated accordingly, right.

And these all BGP messages which are transmitted, they are sent over TCP connections and based on that as I said the routing tables of the routers, they get updated.

Now, BGP comes in various versions.

Presently version 4 is most common and as I said, it is used to communicate between routers, which are across autonomous systems, which we say inter-AS routing right.

Inter-AS means across two different autonomous systems and as I said for exchange of the messages, it uses TCP with well known port number 179.

So, 179 number is reserved for BGP packet transmission.

Continued

And similar to RIP which we talked about, here also we use a version of distance vector protocol; meaning every router will be sending some information to the other router over this exterior links, telling about the distances with other networks and other places.

Now, there is one difference; in RIP which was an interior protocol here, this distances referred to distance between routers, within the autonomous systems.

But here since, we are talking about other networks which may be outside this autonomous system also; so here there is an option of specifying complete routes, not just the next hop.

Suppose, I want to go from x to y what is the paths to be followed, what is the sequence of routers I should follow, that also can be additionally specified in BGP ok, so that if you want, you can have source routing facilities implemented, if of course, your router supports that.

Now, here we have a pictorial view of what we have said.

You see that there are three autonomous systems as you can see.

There are three AS-1, 2 and 3 and these blue links which connect some of the routers they run the BGP protocol.

So, these blue links will be connecting with this router ok.

Continued

So, you see these will be the boundary routers or boundary gateways.

They will be connecting with the routers of the other networks right.

So, it is between these routers that this BGP protocol will be running.

Now, in contrast, inside a network wherever there are routers also for updating information with respect to a particular autonomous system; they will be using the interior protocols.

Typically, RIP was used earlier, but as I said now the most commonly used is OSPF.

Of course there is a version of BGP, called internal BGP that can also be used as an interior routing protocol.

So, let us talk about the different message types that are there in BGP.

BGP is somewhat similar to what we talked about in OSPF; because in OSPF also, routers where trying to maintain some neighborhood connections and through those neighborhood connections they were either sending hello packets to tell the other person that well I am alive or they were sending some kind of route update packets, some kind of link state advertisement through which state of the link which are changing are sent, so that all the routers can update the information in the routing tables.

Continued

Here, there are somewhat similar commands available in BGP and also there is a message type called open.

Well, open says well if I have one router here in one autonomous system and another router here in another autonomous system, this open message can start or initiate a neighborhood connection between the two routers.

So, when this router comes up, it can send an open message, so that this BGP link between the two routers will get established.

Now both the routers will know well, now we can expect to send and receive BGP messages and responses over this connection which has been established.

This update as the name implies, it sends information about changing routes.

So, they can transmit various kinds of information like about a single route, the route changes, some link goes down, some link comes up.

So, the route can change that information might be sent.

Some new routes which are created maybe a new link is established, some new router connections have been established.

Continued

So, some new routes will come up that kind of information are also shared and of course, some of the paths which may become infeasible.

Because some of the links are too slow or maybe some link has gone down that information can also be sent using these update messages and just like hello message in OSPF here in BGP, there is a message called keep alive.

Keep alive message sent between the neighbors will maintain that connection alive.

Well if you do not say keep alive for a certain amount of time, the neighborhood connection will be automatically terminated.

It means that one of the two parties is not interested in exchanging BGP messages anymore, ok and in case of some error conditions happening in the network; some notification messages maybe sent again using this BGP protocol.

So, these are broadly the various messages I am not going into the details.

For details you can refer to a number of materials which are available.

I am just giving you the overall idea ok.

Continued

These are the four message types supported by BGP.

Now, the two end routers through which BGP establish a connections and exchange messages, these are called peers or peer routers or peer gateways.

So, these peers are the one which participate in sending and receiving BGP messages right.

So, initially when everything starts up at the very beginning, the BGP peers will exchange entire routing table with the neighbors.

Initially when the network comes up.

Suppose, I am one of the peers, I will send my entire routing table information that I have to my neighbors.

Similarly I will be receiving similar information from the others, so that all of the routers can update their routing table, with the latest information that is available right.

But after this initial exchange of the entire routing table, subsequently I shall not be sending the entire table anymore.

Continued

I shall only sending information about the changes, some new routes are getting added, some existing routes are getting modified information like that ok and of course, I can mention this keep alive messages are important.

They are sent periodically, so that both the parties know that well the link is alive and the routers are up and running.

Notification messages also I mentioned, they are sent in response to some error conditions.

As I mentioned that there is a version of BGP called internal BGP or in sometimes in short you call IBGP that can also be used by routers to update their information within the same autonomous system also, just as a substitute of OSPF let us say, fine.

Now, this notification message as I said is used to sense and receive some error conditions.

So, what kind of error conditions?

Some of these are mentioned here.

You see, suppose I have a BGP packet coming ok, BGP packet is a TCP packet.

Continued

So, there will be some header and there will be the body of the packet message.

Now, the first thing is that there can be some error in the header, like this is the BGP header right, now on top of the BGP header, TCP will be including its own header to make it a TCP packet.

So, it may so happen that there is some erroneous information in the header part, some of the fields which are invalid.

So, something might happen during transit, some bits might become changed, 0 may become 1, 1 may become 0.

So, if there is some error in the header, so either in the syntax, some invalid values are there or there is an authentication option.

Also I mentioned some of the packets might get encrypted.

So, if the encryption is somewhere wrong, so that the receiving end cannot decrypt it, they will be reported as message header errors.

Similarly, there can be open message errors, where there is some syntax error in the body of the message, some options you have specified which are not one of the valid options.

Continued

This can again happen due to network errors during transmission right.

Similarly, when the routing table changes are being transmitted through the update messages, there can be some error in those messages also, somewhere some corruption may happen, so that the receiver is unable to understand what it means.

Hold timer expired means the keep alive messages are used to inform that the link is up, the connection is running.

There is a timer which is set every time a keep alive message is reached.

After that if the timer expires, it means that the periodic messages are not coming.

So, now I can close the connection, maybe the other side is not responding.

Sometimes, due to some reason, you can forcibly terminate a connection.

Let us say due to some reason, your network is being brought down due to maintenance and other things, you are forcibly terminating connection with the other peers.

Continued

So, there you send a cease error message to your neighbors saying that well I am requesting to terminate or cease the connection right, but there is no other error as such I am wanting to close the connection.

Broadly speaking, there are three functional procedures that are there in the BGP which is achieved through BGP messages.

One is of course, neighbor acquisition, I had said.

Initially a neighbor connection can be set up and can be maintained through keep alive messages.

Neighbor reachability; periodically we check for the receipt of the keep alive messages, if it is not coming, it may mean that the neighbor is not reachable anymore, there is some network problem and network which this is neighbor reachability.

Network reachability means you see BGP, in BGP we maintain information about other networks; because of some link failure somewhere, maybe one of those networks has become unreachable, I cannot connect to that network.

So, that is a list which the router maintains.

It is a list of networks that it can reach and also the ones which it cannot reach.

Key terms

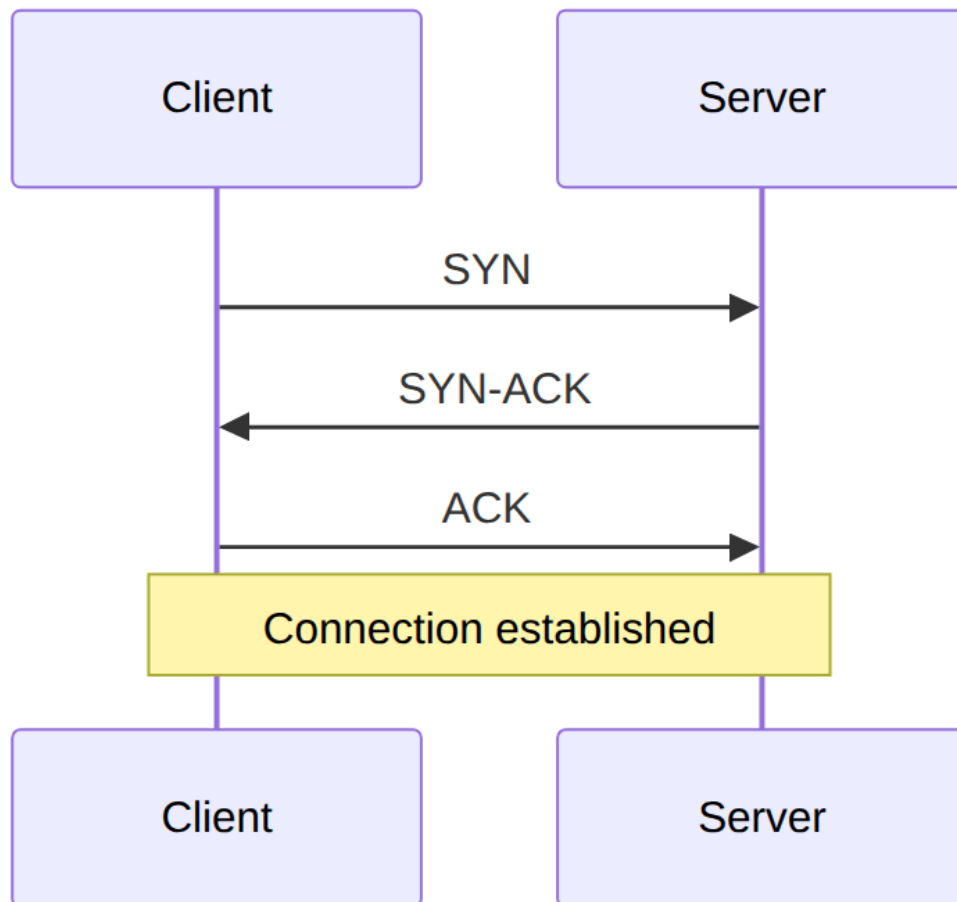


Figure 3: Architecture diagram

Term	Meaning in this lecture
Inter-AS	across two different autonomous systems and as I said for exchange of the messages, it uses TCP with well known port number 179.
Hold timer expired	the keep alive messages are used to inform that the link is up, the connection is running.
After that if the timer expires, it	that the periodic messages are not coming.
Network reachability	you see BGP, in BGP we maintain information about other networks; because of some link failure somewhere, maybe one of those networks has become unreachable, I cannot connect to that network.
The list of network which it can reach	those networks are reachable and the networks, which are not there in the list will mean they are not reachable or I do not have information about them as of now ok.
Mean you are not relying on a, it	other routers to provide you correct information.

Exam-oriented recap

- I shall only sending information about the changes, some new routes are getting added, some existing routes are getting modified information like that ok and of course, I can mention this keep alive messages are important.
- We shall be talking about a new IP version, IP version 6 which is becoming very important in the present day context.
- So, here there is no concept of routing in between.
- So, here if you want, you can use such private networks also.
- So, with this we come to this; we come to the end of this lecture where we give you some very overall idea as to how the border gateway protocol, BGP works as an exterior routing protocol.
- Now, with this our general discussion on routing protocol ends.
- Of course, in the next lecture, we shall be talking about something which is quite related and similar.
- We shall be talking about a new IP version, IP version 6 which is becoming very important in the present day context.

L14: IP Version 6

Transcript: [14-ip-version-6.md](#)

Video: <https://www.youtube.com/watch?v=hy7Cqlbozys>

Week / theme: Week 3 — Routing protocols, IPv6

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- In this lecture we shall be talking about IP Version 6 which is the latest version of the IP protocol.
- So, in this lecture we shall particularly see what are the drawbacks of the older version, IP version 4 and what are the new features this IP version 6 incorporates.
- So, in this lecture we shall be talking about some of the features of IP version 6 protocol, the datagram format in IP version 6 and how certain kinds of address translation are carried out ok.
- So, let us look at the overall scenario first.

What this lecture teaches

Core concepts from the lecture

In this lecture we shall be talking about IP Version 6 which is the latest version of the IP protocol.

Now earlier we have looked at this so called IP version 4 which is more conventional and traditional.

Now IP version 4 has some drawbacks which has lead the designers of this protocols, internet protocols to come up with a newer IP version.

So, in this lecture we shall particularly see what are the drawbacks of the older version, IP version 4 and what are the new features this IP version 6 incorporates.

So, in this lecture we shall be talking about some of the features of IP version 6 protocol, the datagram format in IP version 6 and how certain kinds of address translation are carried out ok.

So, let us look at the overall scenario first.

Now the first thing is that as we have repeatedly said, this TCP/IP forms the backbone of the internet that we see today and IP protocol is most crucial in the sense that all packets that flow through the internet, they are routed using IP protocol.

So, IP ensures that packets reach the intended destination correctly.

Continued

As I had mentioned, this IP version 4 is more widely used today, but there are a certain number of drawbacks in this protocol.

So, once we identify what this drawbacks are, then we can appreciate that what kind of enhancements were needed to enhance the capabilities and that is how this newer version IP version 6 has come.

This is also sometimes known as IP next generation IPNG ok, this both names are used interchangeably.

Now, problems with IP version 4 are as follows.

The first and most important problem is that we are using 32 bit IP addresses, 2^{32} is how much?

Close to 4 billion, but you see with the huge proliferation of devices that are getting connected to the internet today, we are now in an age where we are talking about internet of things, there will be a large number of very simple devices, they will also want to get connected to the internet.

So, how can we give unique IP addresses to everybody, this 4 billion is a rather small number by today's scenario ok.

So, this cannot satisfy your support, the kind of internet growth that has been projected right, this is one.

Continued

Secondly, there are many application, you see there are numerous streaming applications today, streaming audio, streaming video, streaming news and so on and so forth, they need some sort of a real time response.

Like something is happening, you are watching it live, let us say.

Now you will not like, if there is a 10 second pause in between and then the video again starts to play; that is not acceptable in terms of the quality of service right.

So, there are such applications which are increasing day by day where you need real time response.

Real time response means, the delay that we encounter must be within some tolerable limit.

Now what is the tolerable limit, it depends on the application of course, ok.

So, for this kind of real time response one requirement is that you should not change route very frequently, because we mentioned that IP uses some kind of dynamic routing and packets may follow different parts at different times.

Now if we consider the packets that correspond to a video stream, the packets are following one path, suddenly the packet starts following another path which is the longest path, the delay will increase ok.

Continued

So, that kind of variable delay may not be desirable for such real time applications.

Secondly, there are some scenarios that are coming up where you need more complex addressing and routing capabilities; like you recall in IP version 4, you have a 2 level routing; you have a network, you have a host.

Of course, within the host you, forcibly you use the subnet and introduced a third hierarchy, but that is just within the host address space available to you, but IP version 6 allows more number of levels of addressing which becomes more flexible and of course, it will get more complex.

So, main features of IPv6, few of the features are borrowed from IPv4, whatever was there, same thing is carried forward.

First such thing is connectionless, IP version 4 was based on datagrams, IP version 6 is also based on datagrams, which is a connectionless protocol.

So, each datagram are routed independently and each datagram will contain destination address, so that the intermediate routers can take proper and informed routing decisions ok.

And just like IP, if you see in IP, there was a time to live field, it told you what is the maximum number of hops a datagram can take before it will be discarded, at every hop the TTL field was decremented by 1, so whenever it reaches 0, the packets gets discarded.

Similarly, here also the header will contain a similar field maximum number of hops, data datagram is permitted to take, because if you see you have some estimate regarding the maximum number of hops you require to reach a destination.

Continued

If you find that even within that number of hops you are not able to reach, it means there is some problem in the network, may be your packet is following a circular path in a loop, it is got stuck somewhere, so in that case you discard the packet and some of the other general characteristic like fragmentation and so on, those are also retained fine.

Now, let us talk about the new features which are interesting.

The most important feature is that the number of bits in the source and destination IP address is increased, earlier it was 32, now it is 128, which means I can use up to 2^{128} many unique addresses.

Now how large is this; 2^{128} .

This is something you can just, you can actually calculate and find out this is a decimal number which would be having may be 45 digits; 45 digit in decimal number which is huge, but how huge is that?

Just imagine, you think of the surface of the earth, just assume that I want to assign IP addresses to devices on the surface of the earth, 2^{128} is such a large number that on every square meter on the earth surface, you can assign about 6 into 10 to the power 23 unique addresses.

So, this number is indeed huge and it is expected that this will never be too small in the foreseeable future right.

So, this 128 bit is good enough, quite large.

Continued

And another difference is that in IP version 4, there was a header and the header was fixed.

So, whatever packet information the packet carried, it used to be put in the same header, but in IP version 6 what it does, it uses a series of fixed length headers.

Depending on what you need, you add optional headers like, if you do not need fragmentation, you do not use fragment header, if you need then you add a fragment header.

So, there would be multiple headers, which will be connected in a linked list as part of the IP version 6 header, this is the basic concept.

So, a datagram in IP version 6 will consist of a base header that is the basic header which will contain for example, the source and destination addresses and followed by optional extension headers.

There may not be any, but there can be several.

There are some additional features also; like as I had said for real time traffic, there are some additional features which have been included in IP version 6.

Like it is possible for two ends, two hosts to agree on a high quality path, suppose I am viewing streaming video and I tell that well I want a guaranteed bandwidth of 512 kilobytes per second.

Continued

So, they will agree along with the intermediate routers to provide you with a dedicated path.

A path through which you will get dedicated 512 kbps of bandwidth, so that you can receive this streaming media without any disruption right and all datagrams will follow this path.

So, you see this is a slight deviation from the connectionless approach that IP versions is also supports.

For real time traffic, you have something like a connection oriented concept coming in ok.

So, it is like a hybrid now, it is not a pure datagram approach, sometimes when you need to have quality of service guarantee for real time applications, you may also specify the path which the datagrams will follow ok.

And the next thing is that there is some new kinds of addressing which has come up.

Like earlier there were addressing like host to host addressing, you are sending to a particular host or broadcast sent to everybody, but now there is a new addressing mode called anycast addressing which have been introduced.

Anycast means there are a set of computers, I am telling you send this message to anyone of these ok.

Continued

I do not specify which one, but any one of these.

So, if it reaches anyone of these, I am happy.

This is refer to as anycast address and also this provides for dynamic assignments of IP address, it can change over time, this facilitate result there.

So, IP datagram format as I had said, there can be multiple extension headers, the general picture is like this, it starts with the base header which is 40 bytes in size, but there can be multiple extension headers, the size of the extension headers can be different and at the end, you have the data, because typically let us say TCP is running on top and below you have this IP version 4/6 running, IPv6.

So, TCP will be sending you some message for transmission, so that PDU, Protocol Data Unit, transport layer protocol data unit that is the message will be appended with the IP header, whatever there are and they will be send as IP packet.

Now, the base header, 40 bytes, will be having a fixed structure like this, which is some of the fields or somewhat similar to, I mean IP version 4, but because of the change, size of the address as you see, source address and destination address are 128 bits long.

Well 128 bits means how many bytes?

16 bytes, you have 16 bytes here, you have 16 bytes here, version is 4 bits.

Continued

This is the version 6, so this will contain the number 6, 0 1 1 0.

Priority you can specify a priority in the packet now, which in a version 4 is not possible.

Higher priority packet will move faster, router will give higher priority to those.

Flow level, this is again concerned with flow control, some additional fields or information are there.

I am not going into the detail of all of them.

Then comes the payload length which is the size of the total packet.

Next header, I said that there will be a base header and there can be a number of extension headers ok.

Now there will be a field here which will be pointing to the next header, this will be pointing, so this will be like a link list.

Continued

So, this next header field will contain the information that whether there is a, there is an extension header following this or not.

If there is then it will contain what type of extension header and if it does not contain any more, then it will contain some delimiter indicating that there is no more extension headers and this hop limit, I told you about just like time to leave that field is also there.

This is the basic header in IP version 6.

Now talking about the fields I have already mentioned, let me go through this quickly.

This version I told you about, this is a 4 bit field, IP version 6, it will contain the value 6 which in binaries 0110.

Priority; specifies the routing priority class, higher priority packets will be handled faster by the routers.

Flow level; this is particularly used for real time application, where there are applications that require performance guarantee.

So, flow label contain relevant information, this will contain some specific id or code which will allow the packets to follow this particular path for example, right.

Continued

Payload length is the total length of the extension header plus the transport level protocol data unit; that means, the data you are sending.

Next header; as I said, it is the pointer to the next extension header, it identifies the type of information that immediately follows the current header, it can be an IP extension or it can be the actual PDU, TCP or UDP ; that means, the data.

Hop limit; similar to IP version 4, it is decremented by 1 at each hop and the addresses are 128 bits or 16 bytes or octets ok.

So, pictorially it will look something like this.

This is one example, suppose this is a IPv6 packet which does not contain any extension header.

So, in the base header, the relevant fields are there and the next header field contains TCP, which means it will know that the, that whatever follows the base header is the actual TCP PDU which is carrying the data to be transmitted.

Take this next example where there one extension header.

So, base header will contain an information next equal to route, it tells, it has an extension header which gives you some information about route, maybe it is a source routing or specifying the route to be followed.

Continued

And in the next header, this extension header, there will also be a next field which will tell TCP; that means, this will be the actual TCP data.

So, here there can be multiple such header, so this next pointer will be following this headers one by one ok.

Now, the important kinds of extension headers which are there, they are summarized here, there can be a routing header as your seeing here.

Routing here, routing header permits source routing.

No I am repeating here, what source routing actually means.

Now in the conventional packet routing what happens?

There are routers, a packet arrives at a router, the router looks at the destination address, looks at its own routing table and makes a decisions where to forward this packet.

So, it finds out which is the next router I have to send this packet, so that it can reach the destination correctly or if it is destined to a host in my same network, I can directly send it to the network, to that host, directly send it to that host.

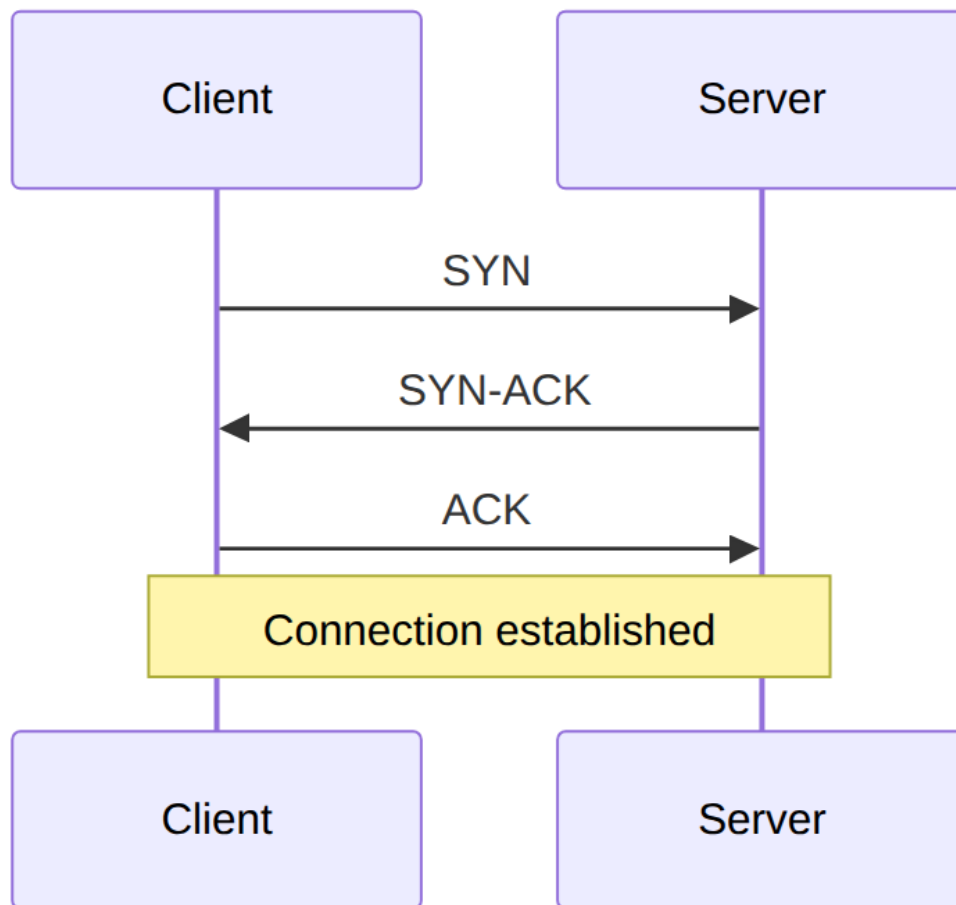


Figure 4: Architecture diagram

Key terms

Term	Meaning in this lecture
Anycast	there are a set of computers, I am telling you send this message to anyone of these ok.
Well 128 bits	how many bytes?

Term	Meaning in this lecture
Well authentication	suppose I am a router, I have received the packet from some other router x.
Multicast or broadcast, multicast	you are sending the packet to all members of a given set, you can say that I am sending a packet to a subnetwork, it should go to all the computers inside that subnetwork, that is a multicast address right.
Only 0	this is all 0, 16 0's, 2D means there are two hexadecimal 0's before that, so like this you can specify.
Now what this	it may depend on from I mean its one place to another, like at the top level it may indicate country for example, that which country the network belongs to, next level may indicate within that country which internet service provider is providing with the connection, the third level can provide some other hierarchical information; that means, within state level or district level some information.
Similarly, you can have another option which	IP version 4 compatible IP version 6 address.

Exam-oriented recap

- The first and most important problem is that we are using 32 bit IP addresses, 2^{32} is how much?.
- Real time response means, the delay that we encounter must be within some tolerable limit.
- The most important feature is that the number of bits in the source and destination IP address is increased, earlier it was 32, now it is 128, which means I can use up to 2^{128} that many unique addresses.
- So, this number is indeed huge and it is expected that this will never be too small in the foreseeable future right.
- Now, the important kinds of extension headers which are there, they are summarized here, there can be a routing header as you're seeing here.
- I must be sure that whatever I am getting is actually coming from the person or the node which it is claiming as a resource that is authentication right.
- So, it must be delivered to every member of the set.
- Now we are saying that IP version 6 is required, we should migrate to IP version 6 whenever we can.

L15: Routing Examples

Transcript: [15-routing-examples.md](#)

Video: <https://www.youtube.com/watch?v=F9ZsWdmXn0Q>

Week / theme: Week 4 — Lab setup, OSINT, recon tools, Nmap
Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- In the last few lectures, actually we have discussed a number of so called routing algorithms which help an IP packet to traverse from a source to the destination through a number of intermediate routers.
- So, as I have said in this lecture we shall basically be working out some examples that will involve routing table and packet forwarding and in general routing of IP packets.
- Now, you look at this routing table.
- So, if there is a match yes, then we follow the outgoing interface, but if there is no match, we do not find any match anywhere then we will be taking the default route, it will be forwarded to eth3 right.
- First, this row, just look at it the subnet mask says the first three bytes are all 1s, last one is all 0, 255 means all 1's.

What this lecture teaches

Core concepts from the lecture

In the last few lectures, actually we have discussed a number of so called routing algorithms which help an IP packet to traverse from a source to the destination through a number of intermediate routers.

We talked about two classes of inter router protocols; the interior and exterior so called routing protocols which help the routers to update their routing tables in a dynamic way.

Today we shall be looking at some examples, where we shall see that given a certain routing configuration in the form of a routing table, whenever a packet comes with a particular destination address, how the packet is handled, how it is forwarded to the correct outgoing link of that router.

So, the topic of today's lecture is routing examples.

So, as I have said in this lecture we shall basically be working out some examples that will involve routing table and packet forwarding and in general routing of IP packets.

So, let us get started.

The first example that we take here, is something like this.

Let us assume that we have a router, let us say we have a router which has four interfaces.

Continued

There are four links to the router and this interface names we are giving as eth0, eth1, eth2, eth3.

Now this eth is the short form for Ethernet, because typically inside a LAN this interfaces are typically Ethernet networks, so that is why we are giving this names like this eth0 to 3.

Now, you look at this routing table.

Well here I have not shown all the fields, the relevant fields only I have shown, destination IP address or the destination network address, subnet mask and which interface to forward it to.

So, you see there are four entries, there are four destination network or host address.

You see the first two entries refer to network address, it starts, it ends with dot 0, but the last one is a host specific address.

Because it is the host specific address, I am using all bits to check, subnet mask is all 1s, but for the others, subnet mask will tell you how many bits in the address I have to check for the network address and the last bits will be for the host address.

So, let us see, now the question is, there is an IP packets in the first example here, there is an IP packet which comes to the router, this was the router and an IP packet comes here, where the destination address is given by this 128.35.57.16.

Continued

This is the destination IP address.

Now we have to check which entry in this routing table are getting a match.

There may be one or even more than one entry where there can be match as we shall see through this example.

The rule is as follows.

We have the IP address, this we already mentioned earlier.

With the IP address we do a bit by bit AND with the subnet mask whatever is specified.

And whatever you get after doing this ANDing, this we compare, whether it is equal to the network address or the host address which is specified in the network table, routing table.

Here it is this destination field.

Continued

So, if there is a match yes, then we follow the outgoing interface, but if there is no match, we do not find any match anywhere then we will be taking the default route, it will be forwarded to eth3 right.

Now this 128.35.15.16 this will be checked with the rows one by one.

First, this row, just look at it the subnet mask says the first three bytes are all 1s, last one is all 0, 255 means all 1's.

So, if you do a bit by bit ANDing with this, you simply get 128.35.57, last 8 bits will become 0, because you are ANDing with eight 0's at the end.

Now if you compare this with the destination address out here, you see that there is a match right, this is exactly the same as this.

So, in the first row you get a match, but in the routing table you do not stop here.

You also check the other rows, I will tell you why, first row there is a match all right.

Now you similarly go to the second row, do a similar checking.

Continued

Now you look at the subnet mask first three are all 255 alright, last one is 128. This 128 means a single one in binary followed by seven 0's.

Now this will be ended with the last byte or octet, this is 16 ok.

So, the first three are all right, first three will be the same, if we ANDed with 255.255.255 it will be 128.35.57, but the last one, the 16 will get ANDed with 128. If you do bit by ANDing you see all are 0s.

So, here again you will be getting the same thing; 128.35.15.0.

So, here also there is a match, there is a match in both the entries in the table, but for the third one, there will be no match as you can see, you are checking for all the bits, so the entire thing remains and this is certainly not the same as 192.0, so it will, there will no match.

So, now you have a situation where there are two matches in the routing table.

Now, when there are multiple matches in the routing table, the rule that is followed is, there is something called longest prefix match, longest prefix match, this is what you look for.

You see in the first row there was a match, but how many bits was checked for the network address; first 8, 8, 8, 24 bits.

Continued

So, the network part was 24 bits, but for the second subnet mask, there were 24 plus in the last one, there was a single one, so it is 25, 25.

So, you will always take the highest one, the longest network number for which a match is found.

So, it will be considered that for the second entry the match is found and this packet will finally, be forwarded to eth1 interface right.

So, this is how forwarding will take place for the first one.

Let us look for the second one now.

Well for the second one; obviously, the first two will not give match, because you are ANDing with 255.255.255.

So, 192.112.17 will be getting anyway, but the destination starts with 128.

So, there will be no match here, there will be no match here, so let us look at the third entry.

Continued

So, you are checking for all the bits; that means, you are ANDing with all 1s, if you AND with all 1s, the same thing will remain, there will be no change 192.112.17.10.

So, now, if you compare it with this destination address also it is not matching, here this 25, here this 10, so here also there is no match.

So, in that case, you will have to take the default route and this packet will be forwarded to eth3.

This is how the packets forwarding will take place right ok.

So, let us move on to the next example, this is a very similar example.

Here I have assumed that the routing table is given, here there are five entries and here again an IP packet is coming whose destination address is 144.16.68.131.

Let us look one by one where the matches are coming, let us take the first one.

The subnet mask is 255.255.0.0, the last two bytes are all 0, so the last two bytes will become 0.

Continued

So, for this one, so after this bit by bit ANDing you will be getting 144.16.0.0 which is actually matching with this, so for the first one you will be getting a match.

Let us look at the second one.

For the second one, you see the third byte of the mask is 224. Now actually what is 224?

224 is nothing, but 111 followed by five 0s, this is 224.

So, you will be doing a bit by bit ANDing of 68 with 224. What is 68?

68 if you convert to binary it will be 01000100, 64 and 4, 68.

So, if you take a bit by bit ANDing, here what you get is, 0.

This 1 and 1 will be 1000000 which in decimal is 64, this bit is 1.

Continued

And the last one is 0, so 0, if we AND with 131, this will be 0 anyway.

So, here also if you do an ANDing, it will become, this will come to 144.16, the third one will become 64, the last one will become 0.

So, you see the destination address is exactly that, so here also there is a match.

So, there is a match in the first two rows.

Let us come to the third row; 255.255.255 all three.

So, the first three 144.16.68 will remain, last one will become 0 which is the same as here.

So, here also there will be a match.

So, there is a match in all three rows of the table.

Continued

The fourth one you can check, here of course, there will be no match, because the first three 255.255.255, 144.16.60, this will remain and this 224 will get ANDed with 131.

Now this if you do 220 say, 131 is what, 131 is 128 plus 3, this is 131.

So, if you do a bit by bit ANDing with 224, 224 is 111 followed by all 0s; so, only the first one will be 1, rest all will become 0s, this is 128, but here it is 64, so it is not matching right.

So, there is a match in this example for the first three rows.

So, now, as I had said in router we follow the longest prefix match rule, you see with respect to the subnet mask, the first row had 16 bits of the network 255.255, second one had 16, 16, and 3, sorry 8, 8 and 3, 19 and the third one had 8, 8 and 8, 24.

So, the third one is the longest prefix.

So, ultimately the match will be identified for the third row and the packet will be forwarded to this eth2 interface right, this is how the packet forwarding will take place.

Now, let us look at a third example where here we are using subnets, sub networks.

Continued

So, you see pictorially the network configuration looks like in the diagram as shown here, this is the router, we are interested in the routing table that we are showing here.

This is the routing table of this router and this router is connected to four sub networks a, b, c, d through these four links small a, small b, small c and small d, and there is an exterior router which is connected to the outside world.

There is another fifth link e which is connected to that exterior router, this is how the connections are.

Now we are assuming that the subnet, sub network addresses, the network addresses are as follows; this, this, this and this.

They all start with two 215.1.2, the last eight bits are sub networked.

So, you see 0 and 64 is the first one, it starts with the last byte 0, it continues till 63, b similarly it starts with 64 and continues till 127, the third one starts with 128 and it will continue till 191 and the fourth one will start with 192, it will go till 255.

Now here suppose some packets are coming.

Let us go to the next slide.

Continued

So, here we will work out this examples for these four packets, these are the destination addresses and we will try to find out what will happen.

In the first one the last byte is 33, 78, 144 and 200.

Now, you see logically speaking you can, now you can identify that said the first one will belong to the range of sub network a I mentioned 0 to 63, second one will be in b, third one will be in c, third one, fourth one will be in d, but let us see how according to this ANDing and comparison this decision is validated.

So, the first IP address which is coming, it has an address 215.1.2.22.

Now, you see let us make a check in this routing table one by one, if a subnet mask of all of them are 192 in the last one.

Now what is 192 ? The first two bits one, rest all are 0.

Now this 33 we will have to AND with this, what is 33?

33 is 0010 and 0001, 32 plus 1. If we do a bit by bit AND, it will become 0000000, all 0s.

Key terms

Term	Meaning in this lecture
This 128 All 0s as the address	a single one in binary followed by seven 0's. it is the default route.

Exam-oriented recap

- So, you will always take the highest one, the longest network number for which a match is found.
- Now all these routing techniques and methods will be very important for you in understanding later on how the actually network based attacks can be mounted or can be taken place.
- Most of the attacks that we talk about, they are mounted through this IP via some routers.
- So, with this I come to the end of this lecture.
- Now here in this lecture and over the past few lectures I have talked about the routing techniques and we have also seen some examples of routing.

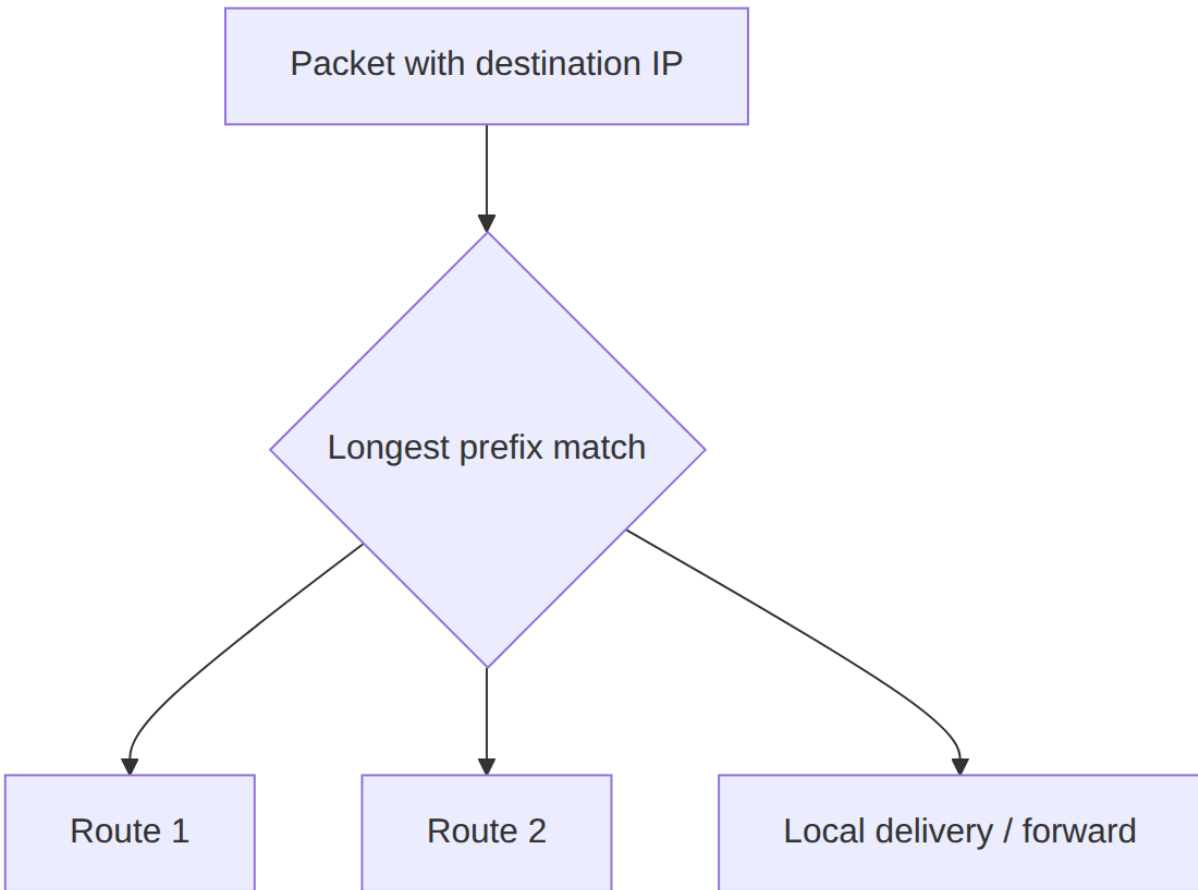


Figure 5: Architecture diagram

- Now all these routing techniques and methods will be very important for you in understanding later on how the actually network based attacks can be mounted or can be taken place.
 - Most of the attacks that we talk about, they are mounted through this IP via some routers.
 - So, once a hacker gets hold of a router, if the routing table gets modified then packets can be routed arbitrarily as per the wish of the hacker ok.
-

L16: Demonstration — Part I

Transcript: [16-demonstration-part-i.md](#)

Video: <https://www.youtube.com/watch?v=nWJ9zyZ486U>

Week / theme: Week 4 — Lab setup, OSINT, recon tools, Nmap

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- So, in today's session we will basically show how to setup your lab environment.
- Now, in this session we will show how to install Kali Linux in VirtualBox.
- So, in today's session we will basically show how to setup your lab environment.
- And basically preferably we use Kali Linux as the attacker machine; that means, the hackers machine.
- So, it is very difficult to arrange so many systems to run all this operating system.

What this lecture teaches

Core concepts from the lecture

Before going to start the demonstration we need to setup our lab.

So, in today's session we will basically show how to setup your lab environment.

So, in ethical hacking we basically exploit different kind of operating system like starting from the older operating system Windows, XP maybe, Windows 7, Windows 8, Windows 10 and some other Linux operating system are also used as a target wise.

And basically preferably we use Kali Linux as the attacker machine; that means, the hackers machine.

So, it is very difficult to arrange so many systems to run all this operating system.

So, that is why we use the virtualization to install all these operating system in a single machine.

So, there are different tools are available for virtual platform like VMware workstation, VMware player or a VirtualBox.

So, VirtualBox is basically free open source software.

Continued

So, that is why we use VirtualBox to create our own lab environment.

So, here is the VirtualBox software.

You can also go to the official website of virtualbox dot org and from there you can download the VirtualBox for different operating system like windows, OS X, Linux, etc.

So, now I already downloaded it from virtualbox dot org.

Now, execute this and run.

Create start menu entry and also want to create a shortcut on the desktop and also create a shortcut in the quick launch bar and register file associations.

It basically takes some time to install Oracle VM VirtualBox 6.0.8.

So, Oracle VM VirtualBox is already installed into our system.

Continued

Now, in the next tutorial, I will show how to install Kali Linux in this virtual platform.

Now, in this session we will show how to install Kali Linux in VirtualBox.

So, this is the VirtualBox platform.

So, first go to the option new and put the name I am going to install Kali Linux version 2019.2 and this is the folder you can also change this folder and it is a Linux operating system and it is Debian 64.

Now, memory size you can also check these things and change it to any other memory size.

Recommended memory size is 2 GB that is 2048 MB.

Then create a virtual hard disk now.

Now, VDI, VirtualBox Disk Image.

Continued

Now, we dynamically allocate the storage on physical hard disk.

Go to next, and file location and size.

This is the file location and this is the size.

8 GB is recommended you can also change it.

So, I will change it to 20 GB.

So, it will create the VirtualBox.

Now, go to the settings, and we need to add the file of that particular operating system.

Go to storage and this is empty.

Continued

Now, choose the virtual optical disk file.

Now, we choose the Kali Linux 2019.2 ISO file which is available in the website of Kali, that is Kali dot org and go to the download Kali Linux and you can get the all the available version of Kali Linux.

I install Kali Linux 64 bit, so these one.

And the version is 2019.2.

This is the latest version.

Now, see I already downloaded it and select this ISO file of Kali Linux 2019.2, then open, now ok.

Now, start the machine, and say it will start the installation.

Now, you need to control this using keyboard.

Continued

Now, go to the install part.

Suppose, I select English and location United States and configure the keyboard as American English.

You can also select your own choice.

Hostname, you can also change the hostname.

I simply continue, and root password, then continue.

Then reenter and continue again.

Now, configure the clock.

I choose the default one.

Continued

I choose the guided use entire disk.

All file in one partition.

It is recommended for new users.

Now, finish partitioning and write changes to disk.

Now, it will take some time to install the system.

A configure the package manager; use a network mirror, I select no.

Now, it is installing grub boot loader.

It is very important.

Continued

So, install the grub boot loader on a hard disk, always yes.

Now, for the first time not manually use slash dv slash hda.

Now, it is finishing the installation.

Finish the installation, continue.

Now, Kali Linux is already installed and it opening now.

So, now use the username root and password which we which we provide at the time of installation.

Yes, now there is the Kali Linux.

We successfully installed Kali Linux into the virtual platform.

Continued

Now, always remember that before going to install any operating system into virtual platform, always enable the virtualization option in your BIOS setup.

Now, this way you can install other operating system such as windows XP, windows 7, 8, 10, and other Linux operating system also and this way you can set up your own lab environment.

Tools mentioned

- **Kali Linux**
- **VirtualBox**
- **virtualbox**
- **VMware**

Exam-oriented recap

- And basically preferably we use Kali Linux as the attacker machine; that means, the hackers machine.
- So, install the grub boot loader on a hard disk, always yes.
- Now, always remember that before going to install any operating system into virtual platform, always enable the virtualization option in your BIOS setup.
- Now, Kali Linux is already installed and it opening now.
- So, now use the username root and password which we which we provide at the time of installation.
- Yes, now there is the Kali Linux.
- We successfully installed Kali Linux into the virtual platform.
- Now, always remember that before going to install any operating system into virtual platform, always enable the virtualization option in your BIOS setup.

L17: Demonstration — Part II

Transcript: [17-demonstration-part-ii.md](#)

Video: <https://www.youtube.com/watch?v=eFpA1W7zULE>

Week / theme: Week 4 — Lab setup, OSINT, recon tools, Nmap

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- Now, we will discuss about first phase of hacking reconnaissance.
- Now, we discuss about some basic operators in Google search.
- Please note that most of the attacks described in the lectures are illegal.
- So, only try your own network and web application.
- It is better if you would rather disconnect your machine from internet.

What this lecture teaches

Core concepts from the lecture

In this course the lab exercise or demo will be attempt only inside our internal network and web application.

Please note that most of the attacks described in the lectures are illegal.

So, only try your own network and web application.

It is better if you would rather disconnect your machine from internet.

NPTEL will not be responsible for any action performed outside your own lab environment.

Now, we will discuss about first phase of hacking reconnaissance.

Reconnaissance is the phase where the attacker gathers information about the target using the active path, passive needs, passive information gathered.

In passive information gathering, we gather a information not find directly communicating with effective.

Continued

We may gather information from any other source, may from search, different search engines, social site or company website.

We can also gather information using open web like WHOIS, Netcraft history of the website like archive.org, maybe sometimes we also use Google search and some search operator in Google like site in url, file type etc.

Now have a look WHOIS database linker.

WHOIS allows us to access information about the target including registration details, IP address, contact information containing the addresses, email id, phone number.

It also displays domain owner and domain register.

Suppose we want to gather information about the domain microsoft dot com.

We already gather information about the registered date, update on, expire on and different name server.

Registered data like name, organization, address, city, state, postal code country even if email id and phone number also.

Continued

Now, using Netcraft we can also gather information.

Netcraft is an internet service company based in England.

Using this service one can find the list of sub domain and the operating system of the corresponding server.

Now suppose using Netcraft tool we are going to gather information about the same domain microsoft dot com.

Now, see we get all the sub domain for that particular domain microsoft dot com and the corresponding OS in which that particular server is running.

History of the website; it is very easy to get a complete history of any website using www dot archive dot org.

Now suppose using the archive dot org we are going to search the history of the website nptel dot ac dot in.

Now, see the particular domain started in the year 2013 and year 2013 there is only one update that is on December 10.

Continued

Now, suppose I want to see the website at that particular date 10th December 2013.

It is just welcome to.

Now suppose I want to see the web application in any other previous date.

Now, see the website is look like this in December 28, 2014.

So, now this way we can gather information from a web application in any previous day.

Google search; the Google search engine is a hacker's best friend especially when it comes to information gathering.

Google supports the use of various search operators which allow user to narrow down and pinpoint search results.

Now, we discuss about some basic operators in Google search.

Continued

Site operator; site operator has been used to limit the result for a particular site.

For example, suppose we are going to limit our search result with only the site microsoft dot com.

So, now I am opening my browser and using the site operator.

I will search for only microsoft dot com domain.

We get only those result which is related with the site microsoft dot com.

See microsoft dot com, microsoft dot com.

So, all the search result is basically related with the domain.

That means with the site microsoft dot com.

Continued

Let us filter those out to see what others sub domains may exist at microsoft dot com.

So, we are going to search site microsoft dot com minus site www dot microsoft dot com.

So, now see this result basically subtract all the result which is related with the sub domain www dot microsoft dot com.

So, you got all the results with the site microsoft dot com other than the sub domain www dot microsoft dot com.

So, see we can get support dot microsoft dot com, todo dot microsoft dot com, support dot microsoft dot com and so on.

Now, we can also use intitle search parameter.

So, using intitle search parameters, search only in those page title for a word or phrase, use exact match for pages.

So, for example suppose I am searching intitle.

Continued

Then with the phrase penetration testing, so we will get all the result where penetration term or penetration testing phrase is basically related.

See penetration testing here is also the term penetration testing is there.

So, by using intitle search parameter we can search a particular word phrase.

Now, we also used inurl search parameter.

So, using inurl search parameter we are basically look for a word or phrase in the document url that can combine with other terms.

Inurl colon then suppose I am searching login term.

See so, using inurl search parameter we search all the login pages.

That means, where the login page is available, it basically search that particular inurl.

Continued

Now, using the search parameter file type we can also search a particular file type like pdf, doc, excel, ppt like this.

So, it basically match only a specific file type.

For example suppose inurl then microsoft dot com and file type is basically pdf.

Now see in the url microsoft dot com we search for the file type pdf.

So, for this particular search operation we get all the pdf file type in the url microsoft dot com.

Here we can also combine two or more search operator to narrow down our search result.

In site microsoft dot com we want to find ppt file with specific phrase penetration testing.

So, our search string is like this.

Continued

Site then microsoft dot com, then file type ppt and then the term penetration testing.

Now see we get all the ppt file for a particular site microsoft dot com related with the term penetration testing.

Now how to use these search parameter to exploit some domain or IP addresses.

I am giving you one example.

Most of the CCTV camera application use the page with default name control slash user image dot html.

So, use the search parameter inurl control slash user image dot html to find out all the CCTV camera facing to the internet with default page control slash user image dot html.

Inurl then colon control slash user image dot html.

Now see we got lots of CCTV camera IP address which is facing to the internet.

Continued

Now some of the CCTV camera are vulnerable.

They may do not have any user id or password.

Sometimes they only use the default user id or password.

Other than these we can also perform maybe brute force attack, maybe dictionary attack for penetrating inside the CCTV camera.

Now suppose I am opening the first link, now see wow.

There is no user id or password.

So, this camera is not secure and it is facing to the internet.

So, we get all the video which basically captured using this particular CCTV camera.

Continued

Now, check the second one.

Wow it is another camera view.

So, this CCTV camera is also not secured.

So, like this we can find out the CCTV camera which is facing to the internet and further we can perform different attack which we will discuss in later tutorial.

Now email harvesting using the harvester tool available in Kali Linux is an email account, username and hostname are sub domain gathering tool.

As an example if you want to find email addresses and hostname for a target domain using Google we can use the tool, the harvester.

Email harvesting, the harvester tool available in Kali Linux is an email account username and hostname or sub domain gathering tool.

As an example if you want to find email address and hostname for a target domain using Google, then we can use the tool the harvester.

Continued

Now from terminal we can use the harvester tool.

Now for help, we can use the harvester dash h.

Now see to specify the domain we need to use dash d option and for data source like Google, linkedin pgp, twitter we need to use dash b option.

Now, suppose I am searching all the mail id or domain name or hostname for nptel dot ac dot in.

So, first we need to use the tool the harvester, then we need to specify the domain name using the parameter dash d, say domain name is nptel dot ac dot in.

Now we need to specify the data source.

So, by using the dash b parameter we can specify the data source and by including the option, all we can include all the data source option.

Now see the tool the harvester search for nptel dot ac dot in, it searching in Google, in pgp.

Tools mentioned

- **Kali Linux**
- **Netcraft**
- **WHOIS**

Commands / syntax from the lecture

- WHOIS database lookup
- WHOIS allows us to access information about the target including registration details,

Key terms

Term	Meaning in this lecture
That	with the site microsoft dot com.

Exam-oriented recap

- Please note that most of the attacks described in the lectures are illegal.
 - Reconnaissance is the phase where the attacker gathers information about the target using the active path, passive needs, passive information gathered.
 - Other than these we can also perform maybe brute force attack, maybe dictionary attack for penetrating inside the CCTV camera.
 - So, like this we can find out the CCTV camera which is facing to the internet and further we can perform different attack which we will discuss in later tutorial.
-

L18: Demonstration — Part III

Transcript: [18-demonstration-part-iii.md](#)

Video: <https://www.youtube.com/watch?v=YaEnPnGkdog>

Week / theme: Week 4 — Lab setup, OSINT, recon tools, Nmap

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- DNS enumeration information such as IP address, server name and often even server functionality can be discovered by DNS enumeration.
- Now, I am discussing some important scanning option; the small s, capital S, this is used for stealth scan.
- In active information gathering, we gather information about domain name system, open port, services, operating system etc.
- DNS enumeration information such as IP address, server name and often even server functionality can be discovered by DNS enumeration.

- We can interact with the DNS server using DNS client such as hosts and DNS lookup tool.

What this lecture teaches

Core concepts from the lecture

Then active reconnaissance, in active information gathering or in active reconnaissance, we gather information by directly communicating with effective.

In active information gathering, we gather information about domain name system, open port, services, operating system etc.

DNS enumeration information such as IP address, server name and often even server functionality can be discovered by DNS enumeration.

We can interact with the DNS server using DNS client such as hosts and DNS lookup tool.

Now, I will use the tool host for DNS enumeration.

So, the command is host, then you need to specify the target by using dash t and for name server we need to use the option ns followed by the domain name, suppose here we are using the domain name microsoft dot com.

So, we caught all the DNS server with the domain microsoft dot com.

Now, mail server enumeration, using mail server enumeration we can find information about all the mail server related with a particular domain, using the host command we can also enumerate the mail server; the command is like this.

Continued

Host then by using dash t we need to specify the mail server mx that is mail exchange server followed by the domain name.

Here we all again use the domain name microsoft dot com, microsoft dot com mail is handled by 10 microsoft dash com dot mail dot protection dot outlook dot com.

Now, suppose we are going to search the mail exchange server for the domain nptel dot ac dot in, host dash t then mx nptel dot ac dot in.

Now, we got all the mail service for the domain nptel dot ac dot in, now DNS zone transfer.

Zone file contained a list of all the DNS name configured for that particular zone, for this reason zone transfer should usually be limited to authorized secondary DNS service only.

Unfortunately, many admin misconfigure that DNS service; as a result anyone asks for a copy of a DNS service zone file and receive it.

Now, suppose I am searching for the DNS server of the domain nptel dot ac dot in, now we got three name server.

Now, suppose I want to transfer the zone for the first DNS server; so, here I also use the command `host -l then the domain name nptel dot ac dot in` followed by the name server.

Continued

So, there is no permission for the zone transfer; so, that is why transfer failed.

Scanning: in scanning the attacker begins to actively gather information from a target machine or network for vulnerabilities that can be exploited.

There are different types of scanning at there like network scan, port scan version or service scan, OS scan, vulnerability scan etc.

Network scan: it basically detect the live host on the network, port scan detect the open port on the host, version a service scan detect the software and the version to the respective service running in any particular port.

OS scan detect operating system, **vulnerability scan** detect computers or computer systems or networks or applications for weakness.

Now, I am discussing some important scanning option; the small s, capital S, this is used for stealth scan.

In this type of scan basically initiate a TCP connection with the target, but never complete the three way handshake that is why this is called stealth scan.

The small s capital T used for TCP connect scan, the TCP connect scan can often the used to gather more information about the target than the stealth scan, as a full TCP connection is made with the targeted host.

Continued

The small s, capital U, UDP scan, the UDP scan access the UDP port on the target system; unlike scanning TCP port, UDP scan expect to receive replies back from system that have tested ports are closed, that is a state or ACK scan.

The ACK scan is used to try to determine if a TCP port is filtered or unfiltered.

Different timing templates are also there T0 that is paranoid, T1 sneaky, T2 polite, T3 normal, T4 aggressive and T5 incent; Nmap offers the simpler approach with 6 timing templates.

You can specify them with the task capital T option and they are number from 0 to 5 or their net.

The template name are paranoid, we use 0 for paranoid, sneaky we use 1, we used 2 for polite, we use 3 for normal, we use 4 for aggressive and we use 5 for insane scan.

The first two are for ideas aversion, polite mode slows down the scan to use less bandwidth and target machine resources, normal mode is the default and so, thus capital T3 does nothing.

Aggressive mode speed scans up by making the assumption that you are on a reasonably fast and reliable network.

Finally, insane mode assumes that you are on an extraordinary fast network or are willing to sacrifice some accuracy for speed.

Continued

This template allows the user to specify how aggressive they wish to be, while leaving Nmap to pick the exact timing values, the templates also make some minor speed adjustment for which fine grained control option do not currently exist.

Now, we are also providing the timing template and their effect and there are different output option are also there.

We used as small o, capital N for normal output.

The normal output option will create a text file that can be used to evaluate the scan result or use as input for other programs, thus small o capital X that is used for XML output.

XML output can be used for input into a number of different applications for further processing or evaluation.

The small o capital G greppable output, greppable output is often used by penetration tester to allow further investigation using tools like great.

But, can also be searched using tools like AWK, ACD and DIFE, thus small o capital S script kiddies output while, not used for CDS penetration testing; the script kiddy output can be fun to use from time to time.

Now, I use the tool Nmap for scanning.

Continued

So, first type of scan is network scan; that means, need to find out all the live host in that network.

So, first we need to use the command Nmap, then we can use some timing option, then to find out all the live host in the network; we need to use the option the small s capital P and then the total range of the network.

Now, finding out the IP address of my machine by using the command ifconfig.

So, the IP address of my machine is 190 dot 168 dot 0 dot 102 and the net mask is 255 dot 255 dot 255 dot 0.

So, I can use the whole network range like this 192 dot 168 dot 0 dot 0 slash 2 55 or I can also use the total subnet like this.

So, Nmap started and it also give us the list of live host in this network.

So, 192 dot 168 dot 0 dot 1 is live and corresponding MAC address is also there, its a Tp link technologies.

So, it is basically the router IP address.

Continued

Now, 192 dot 168 dot 0 dot 100 is also live, then 103 105 106 and 102 so, total 7 hosts are up.

Now, next type of scan that is port scan.

So, for port scan we need to use the option that small p followed by the port number or port name or port range also.

So, suppose I want to scan the port from 1 to 1000, then I need to use the IP address.

So, suppose I am searching means I am scanning the system with the IP address 192 dot 168 0 dot 106 ok.

So, for first 1000 port 999 port are filtered and a single port 135 that is open and the service msrpc is running.

Now, the thing is that what are the filtered, unfiltered, open, closed port.

So, open port basically an port that actively respond to an incoming connection.

Continued

Closed port a closed port is a port and a target that actively responds to a prop, but does not have any service running on the port.

Close port are commonly found on system where no firewall is in place to filter incoming traffic.

Filtered, filtered port are that ports which are typically protected by a firewall of some sort that prevents Nmap from determining whether or not the port is open or closed.

Unfiltered, an unfiltered port is a port that Nmap can access, but is unable to determine whether it is open or closed.

Open or filtered, an open filtered port is a port which Nmap believes to be open or filtered, but cannot determine which exact state the port is actually in.

Closed filtered, a closed filtered port is a port that Nmap believes to be closed or filtered, but cannot determine which respective state the port is actually in.

Now, I am scanning another host for first 1000 port.

So now, the IP address target IP address is 192 dot 168 dot 0 dot 1 ok...

So now, the IP address target IP address is 192 dot 168 dot 0 dot 1 ok.

Port 80 is open for this host and http service is running, next service or version scan; using service or version scan we can find out the exact version of a service which is running in any particular port.

Thus, small s capital V option is basically used for service or version scan, Nmap then timing option dash T capital 4.

Then that is s capital V, then IP address 192 dot 168 dot 0 dot 106.

Port 135 is open and msrpc service is running and version is Microsoft Windows RPC.

Now, I am using different scanning option, suppose I want to scan the first 1000 port, first 1000 port using the stealth scan option.

So, I need to use Nmap then scanning option dash s capital S, then dash capital T4, then the small p followed by the port range and then IP address ok.

So, see here is a difference in scanning in port 80 http service is running, now using TCP scan see what happened.

Continued

So, for TCP scan we need to use the small s capital T option ok, the result would be same.

Now, suppose using the UDP scan we used to scan the port 53.

So, for UDP scan we need to use as small s capital U option and we are going to scan the port 53.

So, p then 53 port number and then the IP address; no it cannot able to find out the port 53 in UDP scan.

I am using UDP scan, I am going to scan another host that is the IP address 192 dot 168 dot 0 dot 106 ok; port 53 UDP port is basically closed.

Now, I am performing ACK scan, for ACK scan we need to use the small s capital A and then suppose I am going to scan first 1000 port for the IP address 192 dot 168 dot 0 dot 106 ok.

See port 139 and port 445 tcp port is basically filtered ok.

So now, I am going to use some output option

So now, I am going to use some output option.

So, suppose I want to store this result in a txt file; so, for txt output we need to use that small o capital option followed by the filename.

Suppose, here the file name is scan dot txt and the location is root plus myfile ok.

Now, go to the location, now this is route and that is my folders my file and then this is the result scan dot txt.

See so, this way we can store the result in a text file and also in xml file and grepable option is also available.

Now, we are, scan is also there to find out the operating system for a particular host.

So, for operating system we need to use the option dash capital O ok; see that is the result Microsoft Windows ok.

So, the victim machine basically run over Microsoft Windows.

Continued

Now, there is also some other scanning option are also available like aggressive scan; Nmap then suppose I am using the timing option T4, then for aggressive scan we can use the option dash capital A and then the IP address of the effective machine.

So, by doing the aggressive scan we can find out the service scan and operating system scan together.

It gives us the result about some common services which is running in particular port and as well as the operating system also.

Now, see PORT 80 is open and TP LINK WR 740N WAP http configuration version are there and 1900 TCP port is open upnp service is running and the version is ipOS upnp and the thing is that operating system is linux with the kernel 2.6.

So, we got all these details by using the aggressive scan, sometimes firewall also block the ping request sent by the tool and map.

So, in that case we can use the option dash capital P small n to bypass the firewall; so, it basically no pings scan.

See that is now packet fragmentation, an Nmap scan will use tiny IP fragments if that dash small f is specified, by default Nmap will include up to 8 bytes of data in each fragment.

So, a typical 20 or 24 byte TCP packet is sent in 3 tiny fragments, every instance of dash small f adds 8 to the maximum fragmented data size; see that is the result.

Tools mentioned

- **Nmap**

Commands / syntax from the lecture

- Nmap offers the simpler approach with 6 timing templates
- Nmap to pick the exact timing values, the templates also make some minor speed adjust
- Nmap for scanning
- Nmap started and it also give us the list of live host in this network
- Nmap from determining whether or not the port is open or closed
- Nmap can access, but is unable to determine whether it is open or closed
- Nmap believes to be open or filtered, but cannot determine which exact state the port
- Nmap believes to be closed or filtered, but cannot determine which respective state t
- Nmap then timing option dash T capital 4

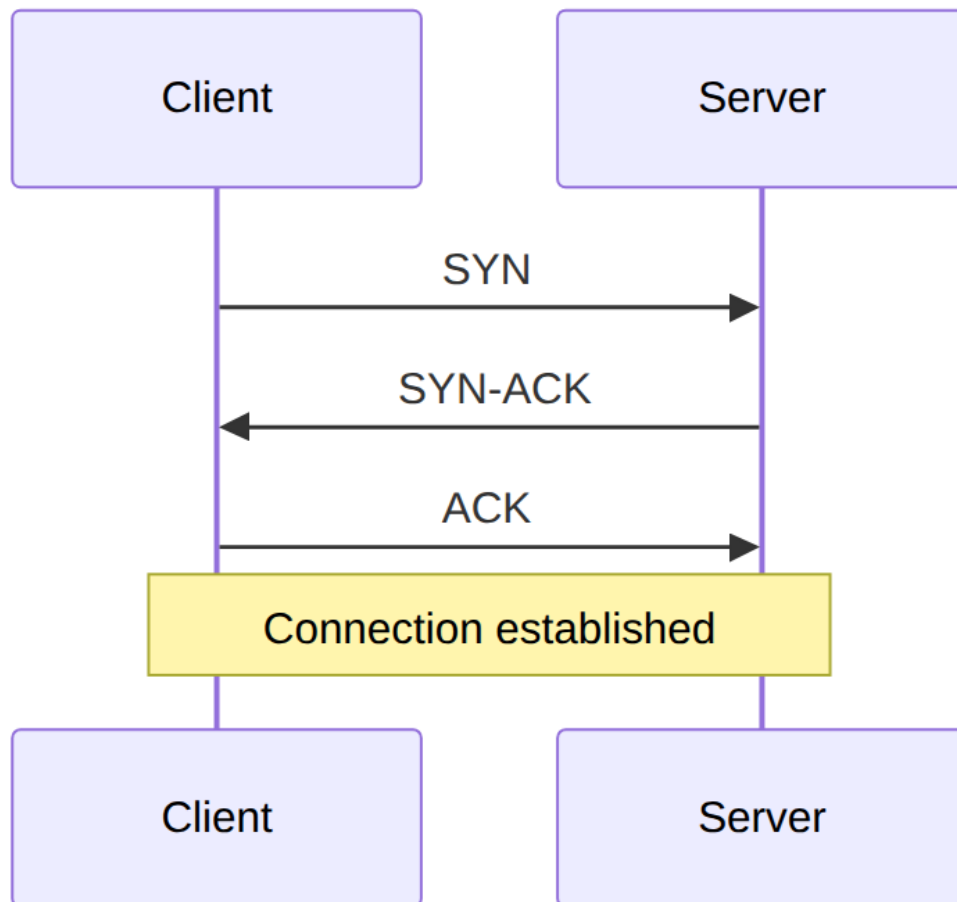


Figure 6: Architecture diagram

- Nmap then scanning option dash s capital S, then dash capital T4, then the small p fo
- Nmap then suppose I am using the timing option T4, then for aggressive scan we can us
- Nmap scan will use tiny IP fragments if that dash small f is specified, by default Nm

Key terms

Term	Meaning in this lecture
So, suppose I am searching	I am scanning the system with the IP address 192 dot 168 0 dot 106 ok.

Exam-oriented recap

- Scanning: in scanning the attacker begins to actively gather information from a target machine or network for vulnerabilities that can be exploited.
- There are different types of scanning at there like network scan, port scan version or service scan, OS scan, vulnerability scan etc.
- OS scan detect operating system, vulnerability scan detect computers or computer systems or networks or applications for weakness.
- Now, I am discussing some important scanning option; the small s, capital S, this is used for stealth scan.
- In this type of scan basically initiate a TCP connection with the target, but never complete the three way handshake that is why this is called stealth scan.

L19: Nessus Installation

Transcript: [19-nessus-installation.md](#)

Video: <https://www.youtube.com/watch?v=8rjEaZP2pnw>

Week / theme: Week 5 — Nessus, system hacking, malware, ARP/MITM

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- So, first we need to install the tool Nessus.
- So, we can download the tool Nessus from the official website of Nessus.
- Now, go to free trial to use the free version of Nessus vulnerability scanner.
- sudo apt install Nessus ok, it start installing ok.
- So, let us start the Nessus service by typing slash etc slash init slash init dot d slash Nessusd start.

What this lecture teaches

Core concepts from the lecture

Now, we use the tool Nessus for vulnerability scanning.

So, first we need to install the tool Nessus.

So, we can download the tool Nessus from the official website of Nessus.

We can use the trial version.

Now, go to free trial to use the free version of Nessus vulnerability scanner.

Try Nessus free for 7 days.

`sudo apt install Nessus` ok, it start installing ok.

So, let us start the Nessus service by typing `sudo systemctl start Nessusd` start.

Continued

So, starting Nessus, next go to `https://kali:8834`, to open the Nessus.

`https://kali`, then it basically use the port number 8834.

So, we need to run port 8834; so, it showing it connection is not secure, go to advance and then go to add exception.

Then get certificate, then confirm security exception, then Nessus will open, then start Nessus essential.

Now, you need to keep your name, first name and last name, now in an ID me exploiter at the rate proton mail dot com.

So, I already have the activation code so, that is why I am skip these steps otherwise you can get your activation code from here in this email ID.

So, for the time being I skip the part and it asking for activation code.

Now, here is my activation code and my mail ID.

Continued

So, copy this and put here now continue.

Now, you need to keep the username .

Now, its downloading plugins, it will take some time to download the plugins; now, its compiling plugins.

Now, we successfully install the tool Nessus.

In the next tutorial, I will show you how to use the tool Nessus to find out the vulnerabilities for a particular system.

Tools mentioned

- Nessus

Exam-oriented recap

- Now, we use the tool Nessus for vulnerability scanning.
 - Now, go to free trial to use the free version of Nessus vulnerability scanner.
 - Then get certificate, then confirm security exception, then Nessus will open, then start Nessus essential.
 - Now, here is my activation code and my mail ID.
 - So, copy this and put here now continue.
 - Now, you need to keep the username .
 - Now, its downloading plugins, it will take some time to download the plugins; now, its compiling plugins.
 - Now, we successfully install the tool Nessus.
-

L20: How to Use Nessus

Transcript: [20-how-to-use-nessus.md](#)

Video: <https://www.youtube.com/watch?v=J3jTrRtV0NQ>

Week / theme: Week 5 — Nessus, system hacking, malware, ARP/MITM

Instructor: Prof. Indranil Sengupta, IIT Kharagpur

Learning objectives

- Now, today's session we will discuss about the vulnerability assessment using the tool Nessus.
- Now, where is all the option for discovery?.
- Discovery option is a ping method, ARP and TCP and it also use the ICMP and maximum number of tries is 2; that means, it try twice for a particular request.
- So, you can choose your own option, then service discovery enumerate all SSL or TLS cipher.

What this lecture teaches

Core concepts from the lecture

Now, today's session we will discuss about the vulnerability assessment using the tool Nessus.

Here is my tool Nessus, first we need to login into the tool Nessus.

So, use the login credential which you use at the time of installation.

So, here is the interface to scan a system, first we need to go to the new scan.

So, there are several options available: advance dynamic scan, advance scan, audit cloud infrastructure, badlock detection, basic network scan; lots of options are there.

So, for the time being we are using advance scan.

So, in these fields first we need to put the name of the scan, suppose the name is NPTEL and description is a Demo, folder My Scans and target, so, in this case my target is sorry target IP address is 192 dot 168 dot 0 dot 1 0 4 and 192 dot 168 dot 0 dot 106.

So, this way we can use multiple IP address for scanning separated by comma.

Continued

Now, where is all the option for discovery?

Discovery option is a ping method, ARP and TCP and it also uses the ICMP and maximum number of tries is 2; that means, it tries twice for a particular request.

Then port scanning part is also there to enumerate the local port to use SSH and WMI, SNMP and only run network port scanner, if local port enumeration failed.

So, you can choose your own option, then service discovery enumerates all SSL or TLS cipher.

Now, assessment part is there and in assessment part Brute force assessment, web application assessment, then Windows assessment and malware assessment.

All options are there, you can choose your own option; for the time being I am of the scan for malware.

Then report we can also use this option to generate the report.

And in advance part there are some options are also there, network timeout; here network timeout is 5 and a maximum simultaneous check per host that is also 5.

Continued

Maximum simultaneous host per scan that is also 5, you can also change your own option.

Now, where presidential is there and plug in part are also there.

So, you already install plug in part which is shown in the tool installation part.

So, all the plug ins are here which can help us to find out the different vulnerabilities right; save this scan.

Now, I am starting the scan by clicking on the button launch, scan already started and it will take some time to complete the whole scan.

And, once we got the scan result and by analyzing that result we can only find out the vulnerabilities and using those vulnerabilities further we can try to penetrate inside the victim machine.

So, let us wait for some time to get the result of vulnerability scanning, now see Nessus complete the scan.

So, let check the result.

Continued

So, you basically scan for 2 systems, one is with the IP address 192 dot 168 dot 0 dot 104.

And, it has several vulnerabilities, total 70 vulnerability is there and out of this 70 vulnerability 9 critical vulnerability is there, 5 high vulnerability is there, 23 medium vulnerability and 8 low vulnerability is there and 129 that all are the information.

And, here is the details of all the vulnerability SSL multiple issues and a bind shell backdoor detection.

A shell is listening on the remote port without any authentication being required, an attacker may use it by connecting to the remote port and sending command directly.

So, using this vulnerability one can attack to the victim machine and some other vulnerabilities also there.

Let us check the other IP address 192 dot 168 dot 0 dot 106, but I have total 21 vulnerability and 1 critical vulnerability is there and 2 medium vulnerability is there and 36 information is there.

And, see MS 17 dash 010 security update for Microsoft Windows SMB server and the remote Windows host is affected by the vulnerability with the CVE number 2017 dash 0143 CVE 2017 dash 0144 CVE 2017 dash 0145 and also CVE 2017 dash 0146 and CVE 2017 dash 0148.

An information disclosure vulnerabilities also exist in this machine and some other vulnerability is also there.

Continued

Like in medium vulnerability MS 17 dash 047 is also there, remote Windows host is affected by an elevation of privilege vulnerability in the SAM and local security authority.

So, all the vulnerability and the possible solution are listed here.

So, this way we can find out all the vulnerability using the tool Nessus and further we use all these vulnerability to penetrate inside the victim machine.

So, this phase is basically call vulnerability assessment phase and in vulnerability assessment phase, we find out all the possible way by which a attacker can penetrate inside the victim machine.

But in the next phase; that means, in penetration testing phase a attacker can try with all these vulnerability to check which vulnerability is working and which one is not; so, that is called penetration testing.

So, in the next week, in next tutorial I will show you how to use all the thing, all the information gathering part which we covered in this week and using all that information how can one penetrate inside the victim machine.

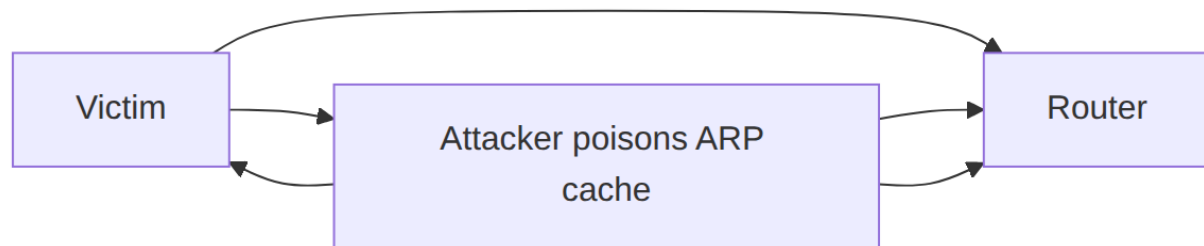


Figure 7: Architecture diagram

Tools mentioned

- **Nessus**

Exam-oriented recap

- Now, today's session we will discuss about the vulnerability assessment using the tool Nessus.
 - So, all the plug in are here which can help us to find out the different vulnerability right; save this scan.
 - So, let us wait for some time to get the result of vulnerability scanning, now see Nessus complete the scan.
 - And, here is the details of all the vulnerability SSL multiple issues and a bind shell backdoor detection.
 - A shell is listening on the remote port without any authentication being required, an attacker may use it by connecting to the remote port and sending command directly.
 - So, using this vulnerability one can attack to the victim machine and some other vulnerabilities also there.
 - Let us check the other IP address 192 dot 168 dot 0 dot 106, but I have total 21 vulnerability and 1 critical vulnerability is there and 2 medium vulnerability is there and 36 information is there.
 - An information disclosure vulnerabilities also exist in this machine and some other vulnerability is also there.
-